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ROYER et al.(10) **Pub. No.: US 2025/0284069 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **TELECOMMUNICATIONS CONNECTOR
WITH LATCH RELEASE MECHANISM**(60) Provisional application No. 63/032,170, filed on May
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G02B 6/38 (2006.01)
(52) **U.S. Cl.**
CPC **G02B 6/3858** (2013.01); **G02B 6/38875**
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Claremont, NC (US)(57) **ABSTRACT**(21) Appl. No.: **19/217,003**(22) Filed: **May 23, 2025****Related U.S. Application Data**(63) Continuation of application No. 18/000,260, filed on
Nov. 29, 2022, filed as application No. PCT/US2021/
034612 on May 27, 2021, now Pat. No. 12,345,925.

The present disclosure relates to a telecommunications connector. The connector includes at least one connector portion including a front housing portion coupled to a rear housing portion, the front housing portion being rotatable relative to the rear housing portion about a longitudinal axis, the front housing portion including a ferrule and a latch, the latch defining front and rear ends and being pivotable relative to the rest of the front housing portion at a connection portion. A boot mounted on the telecommunications connector is movable longitudinally relative to the rear housing portion, wherein the boot is configured to engage at least a portion of the latch of the front housing portion and cause the latch to pivot relative to the rest of the front housing portion as the boot is moved relative to the rear housing portion.

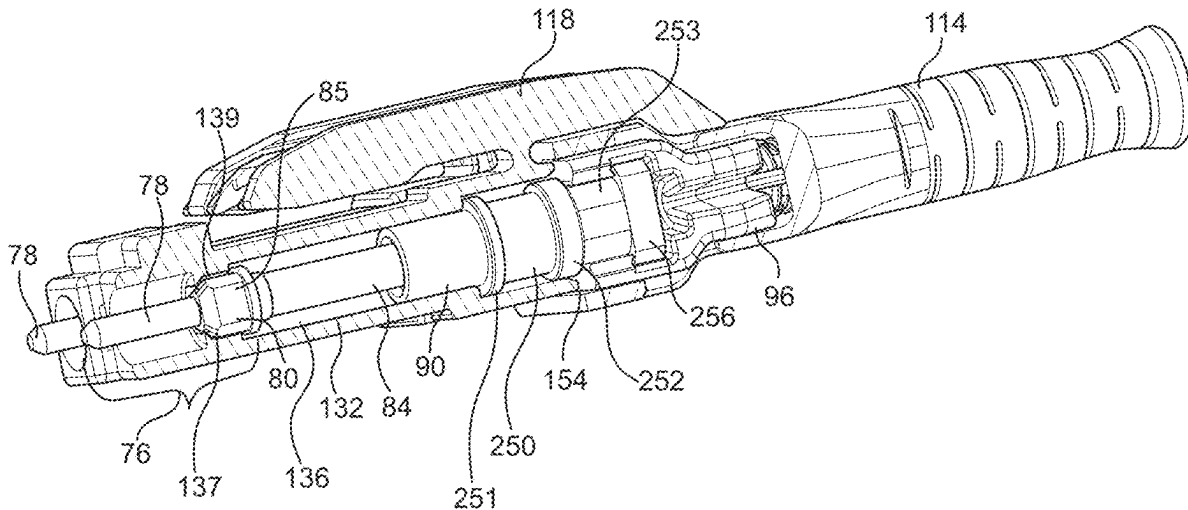


FIG. 1
(PRIOR ART)

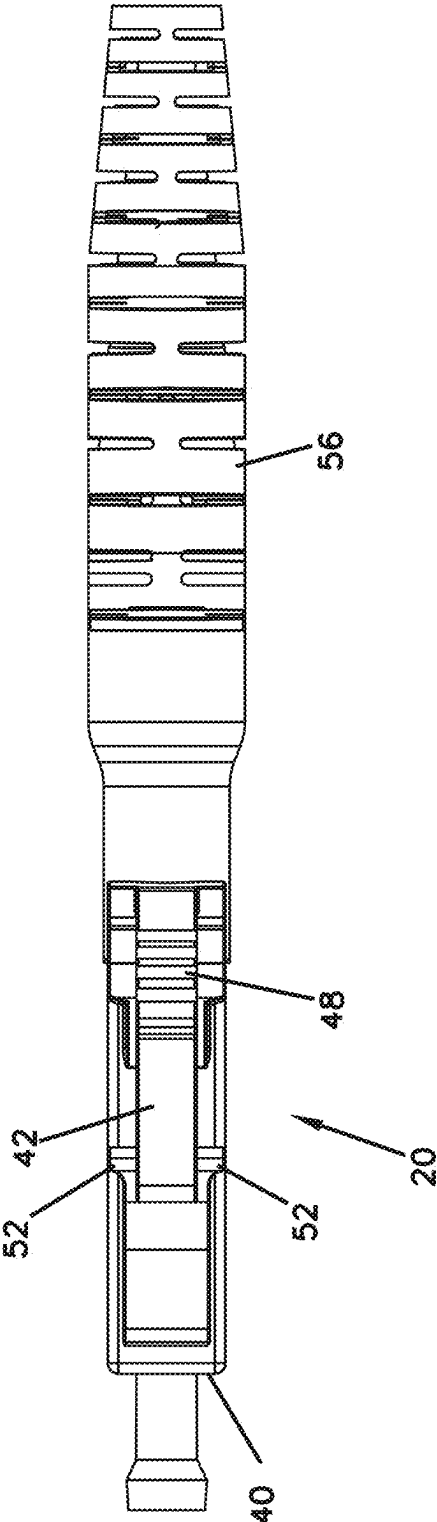


FIG. 2
(PRIOR ART)

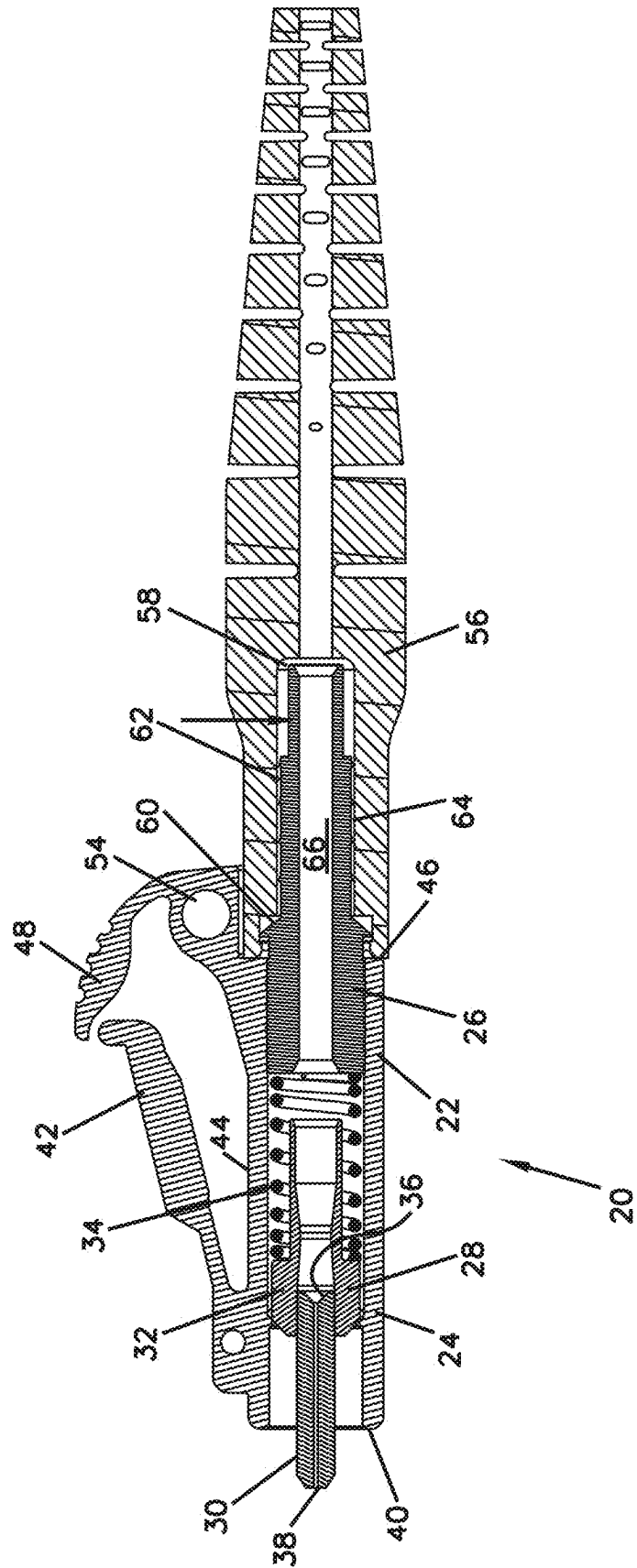


FIG. 3
(Prior Art)

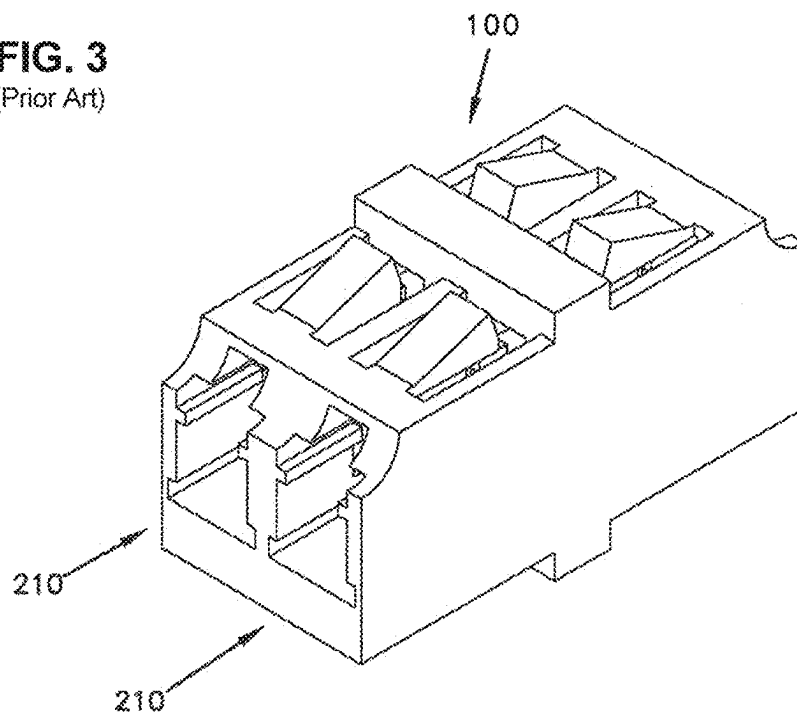
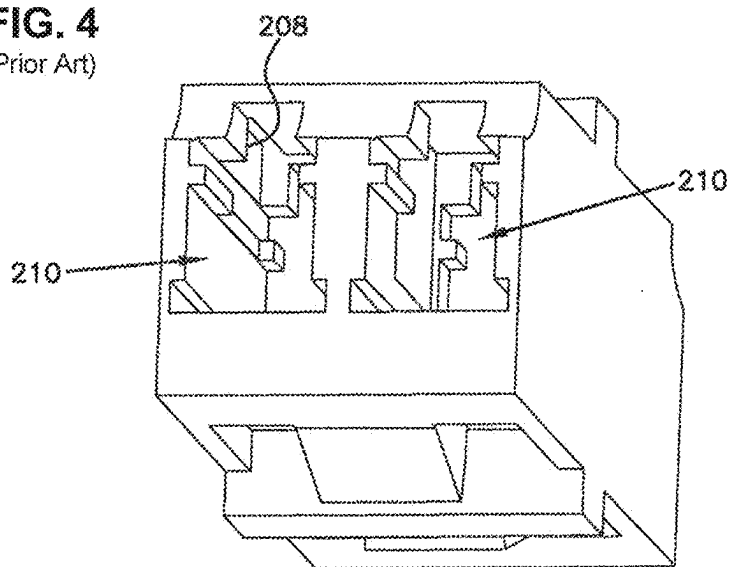


FIG. 4
(Prior Art)



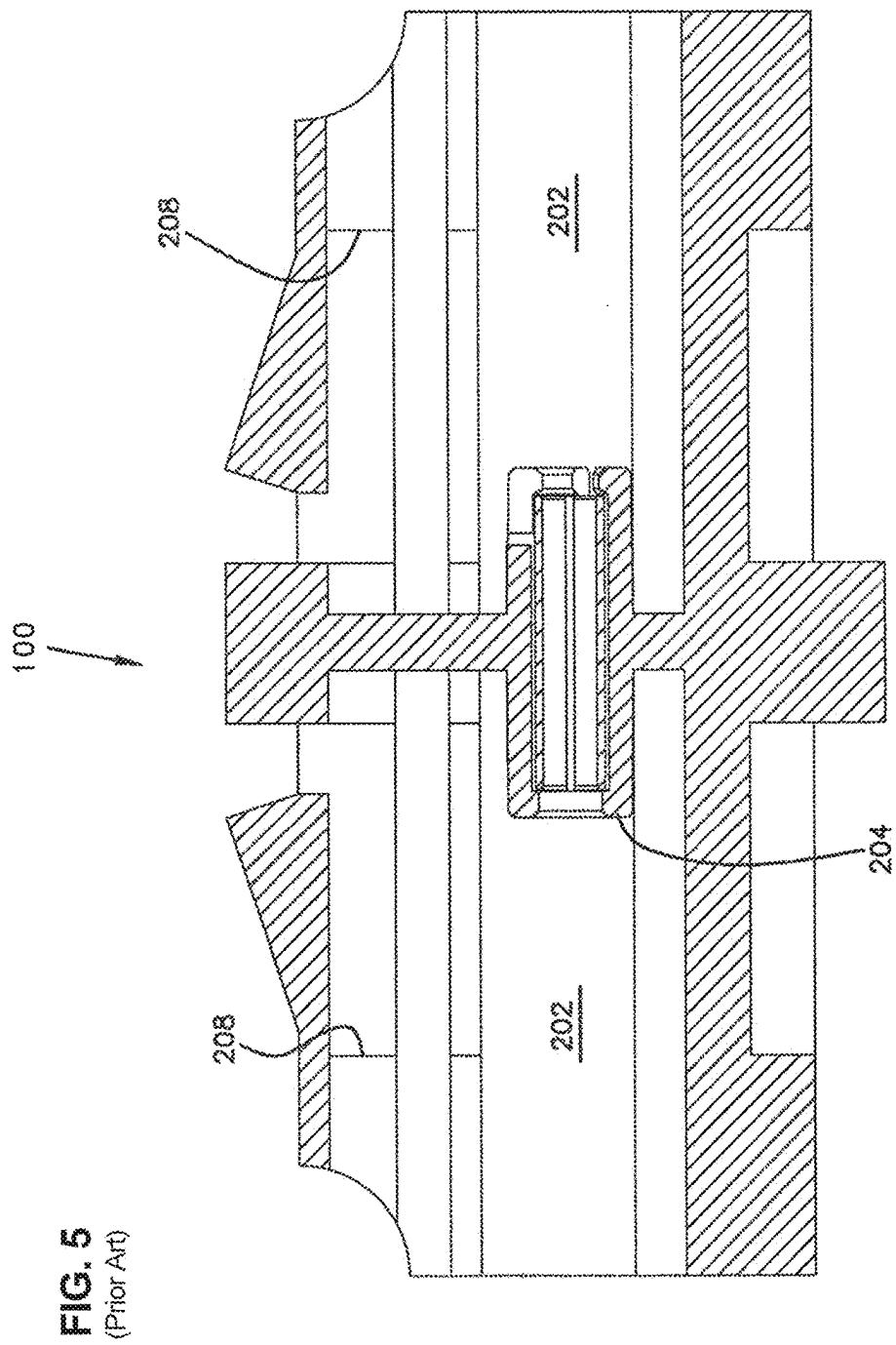


FIG. 6

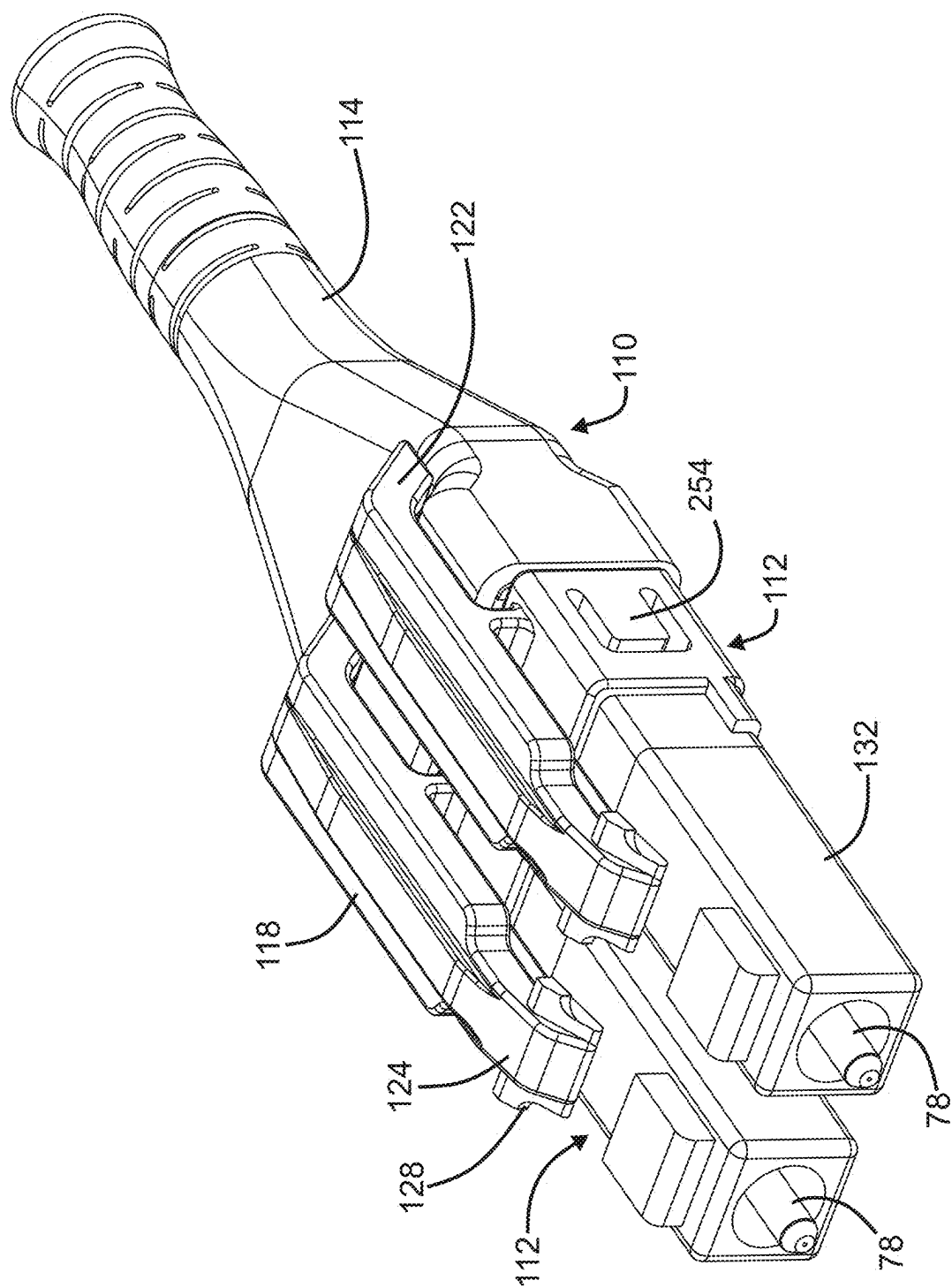


FIG. 7

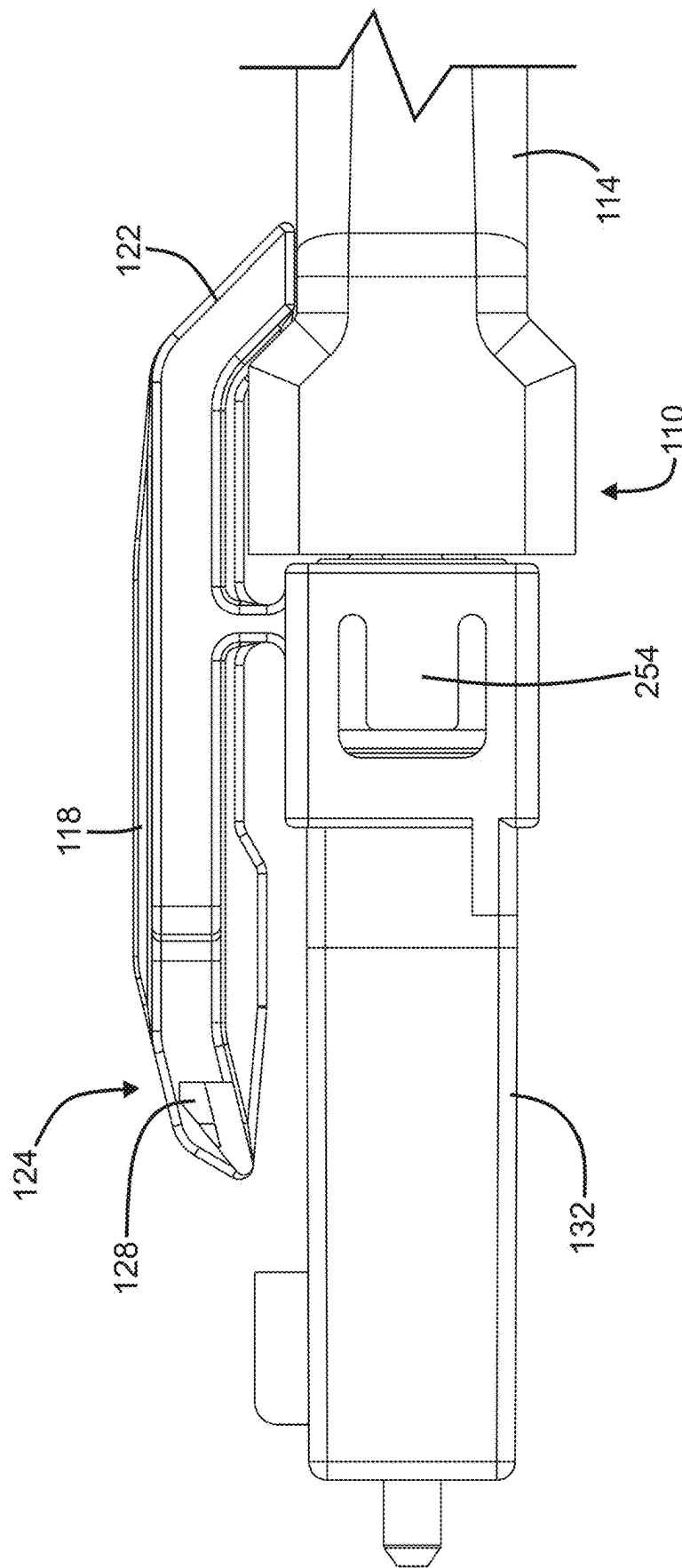
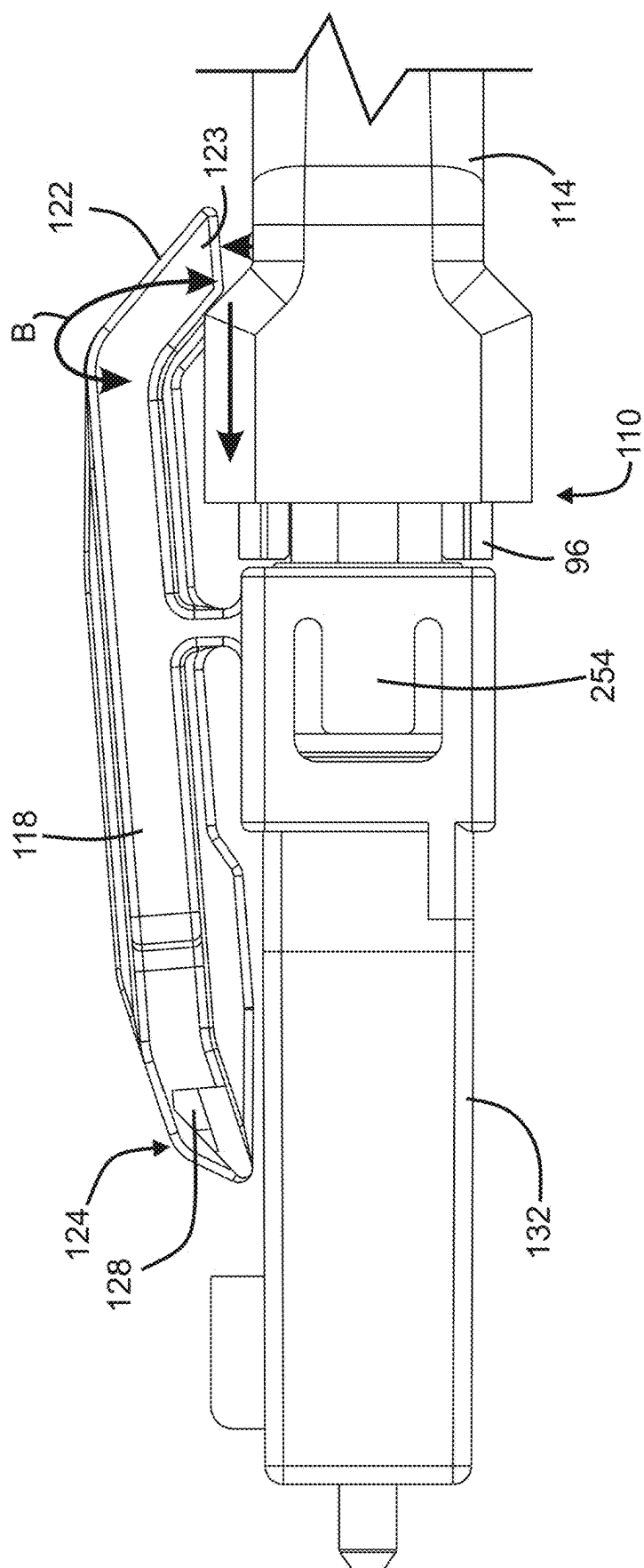


FIG. 8



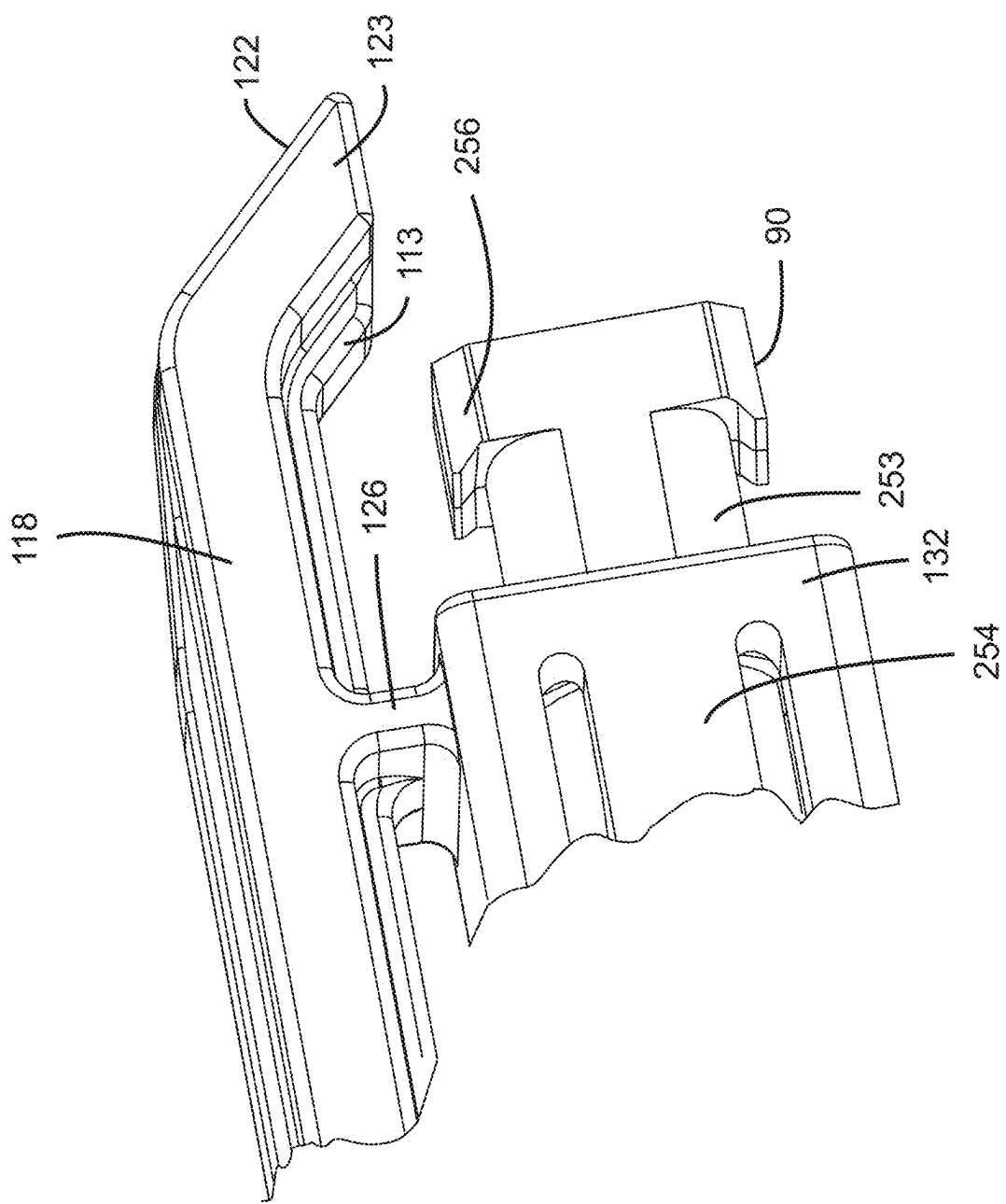
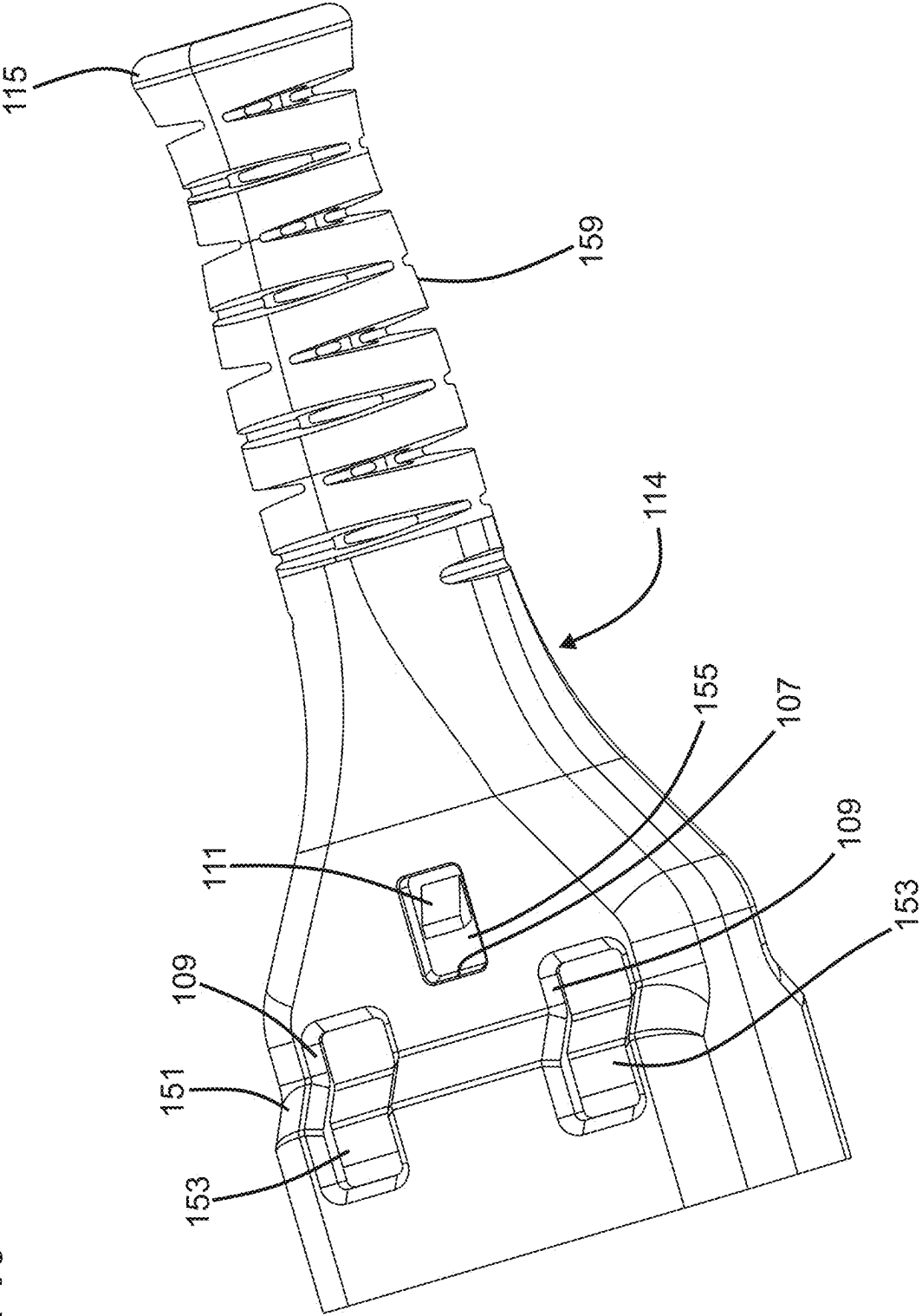


FIG. 9

FIG. 10



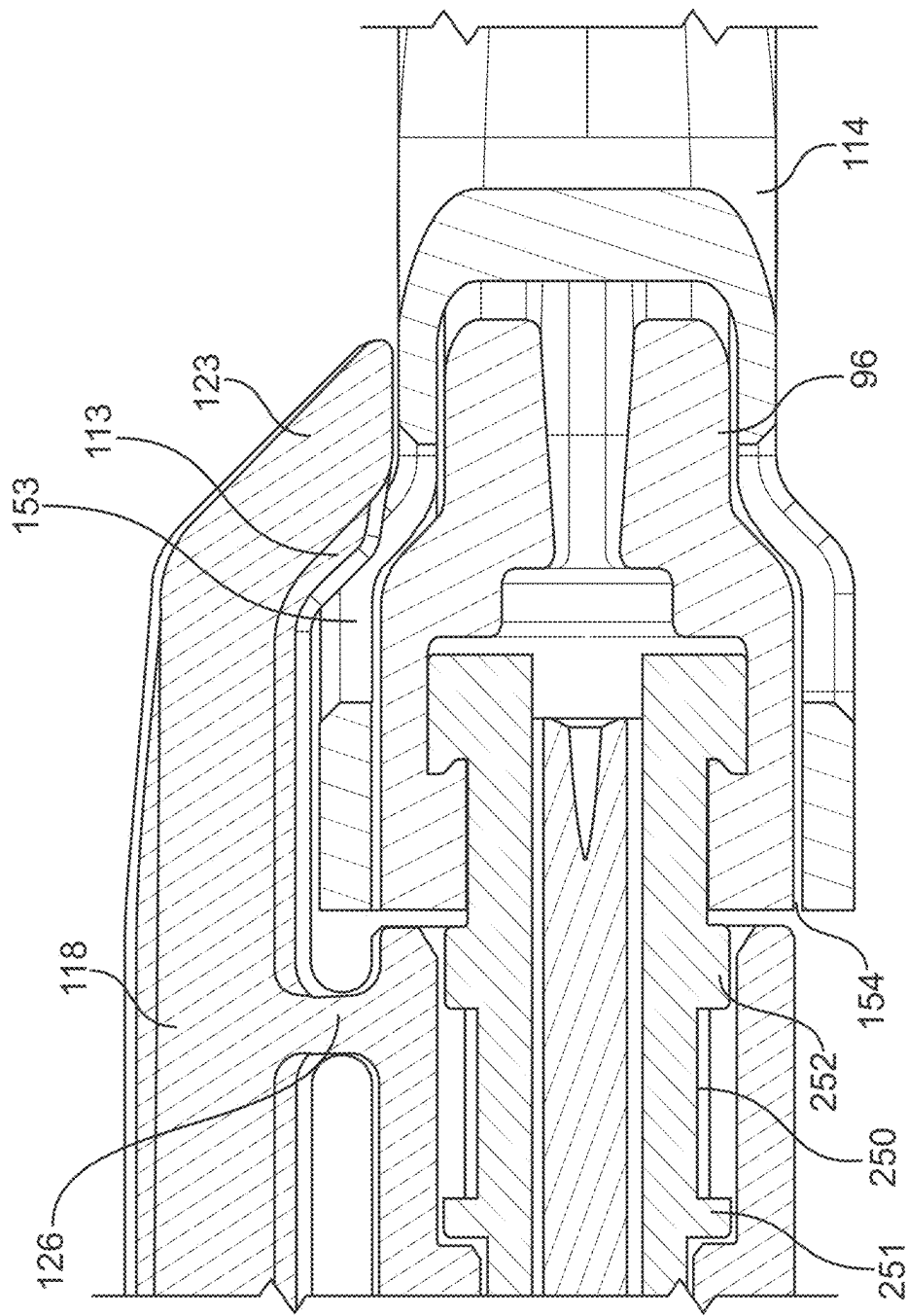
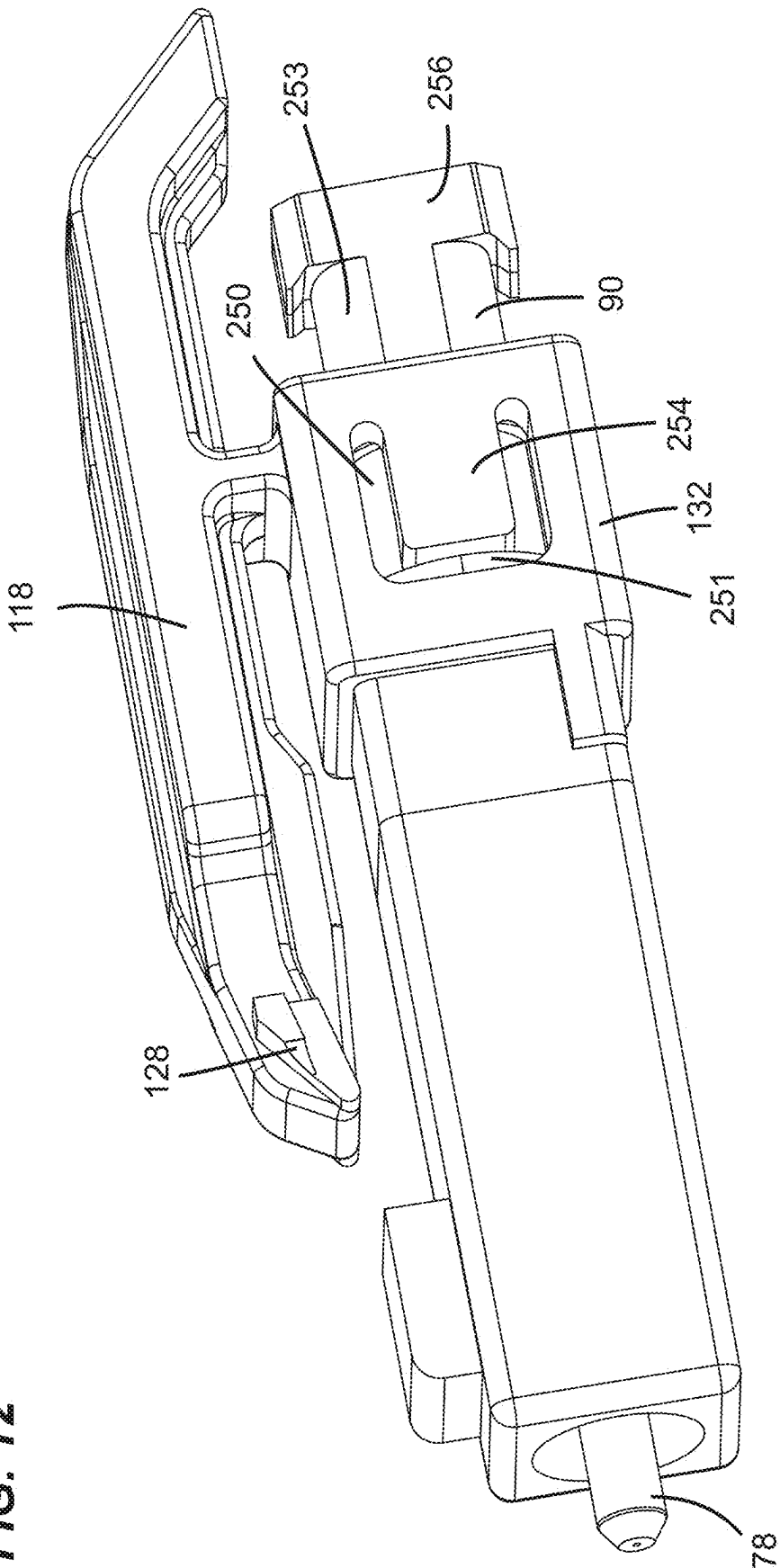


FIG. 11

FIG. 12



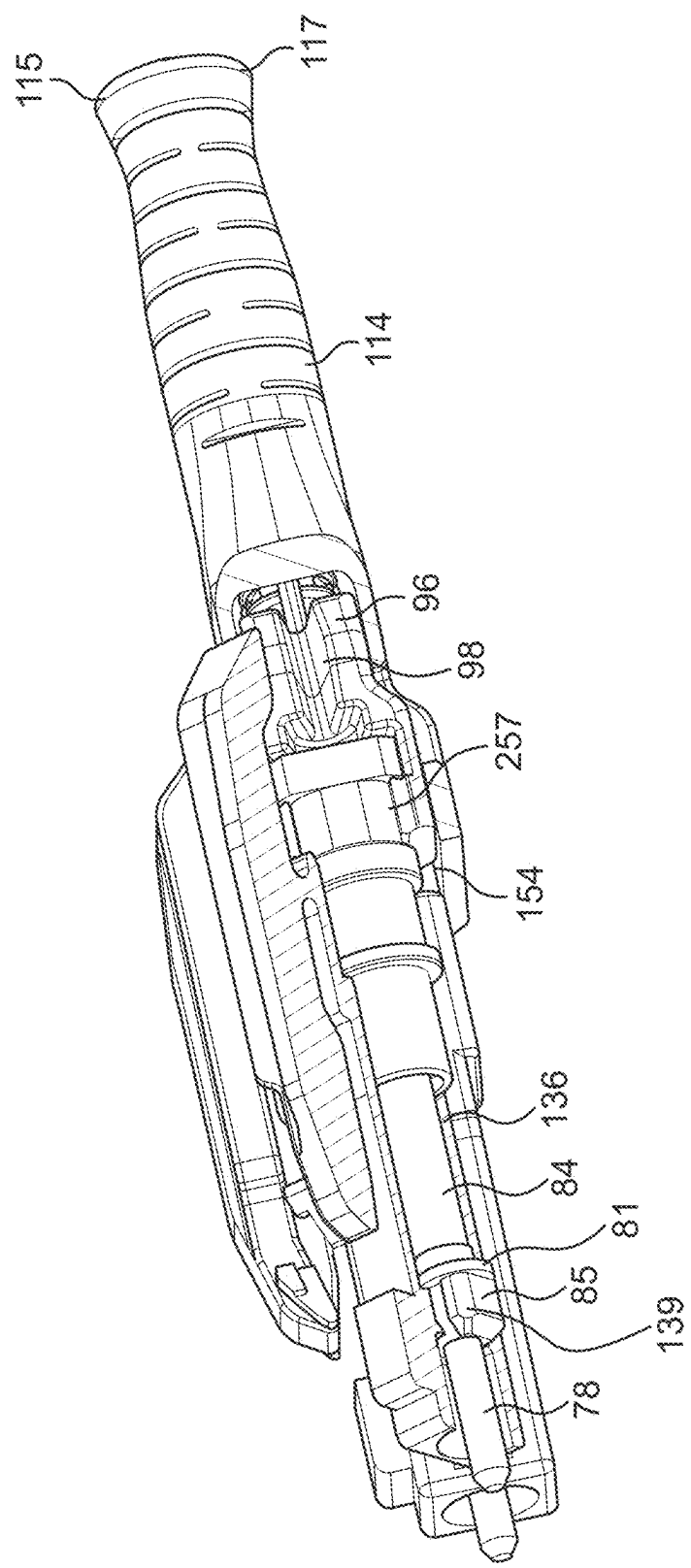


FIG. 14

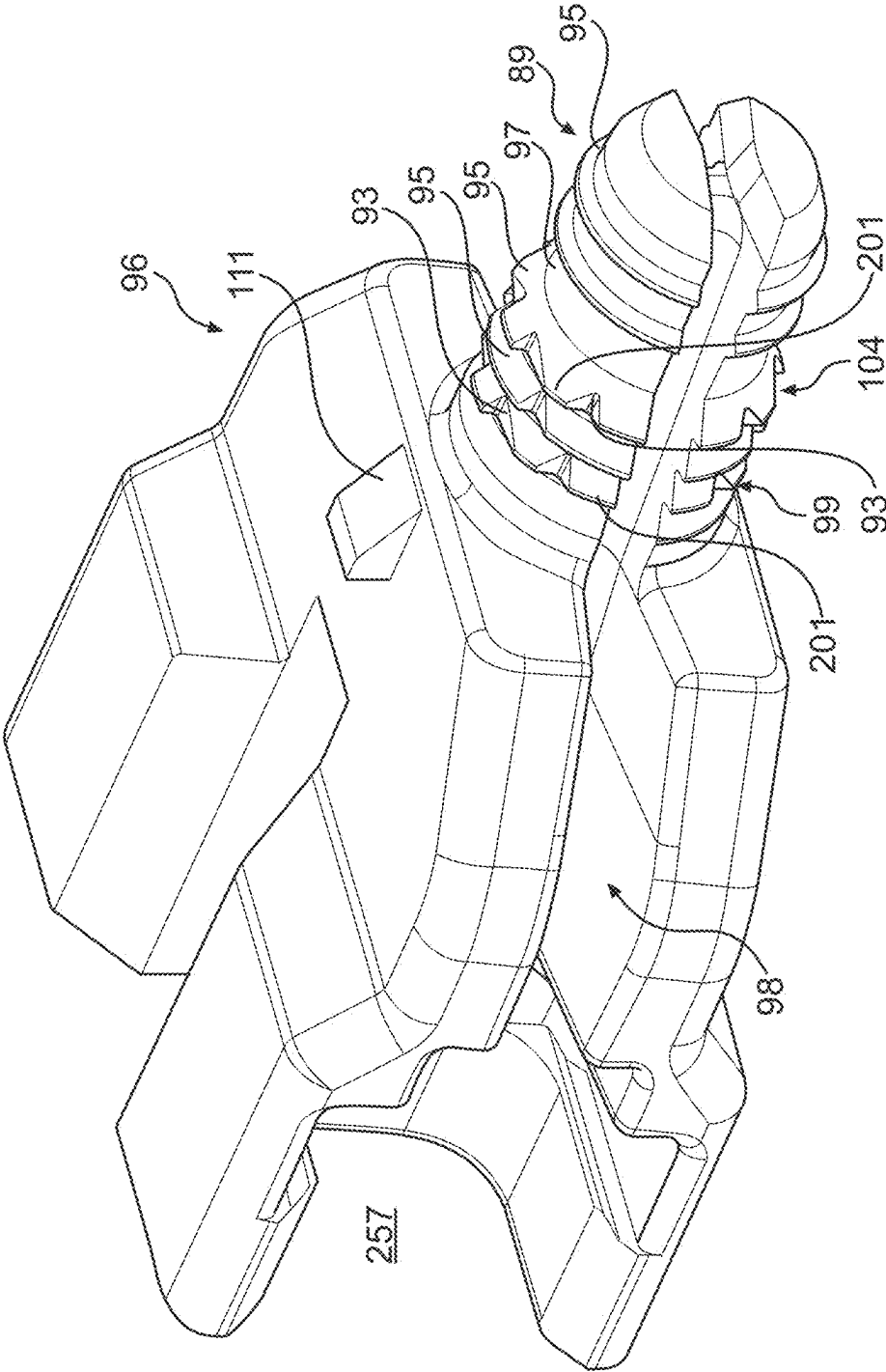


FIG. 15

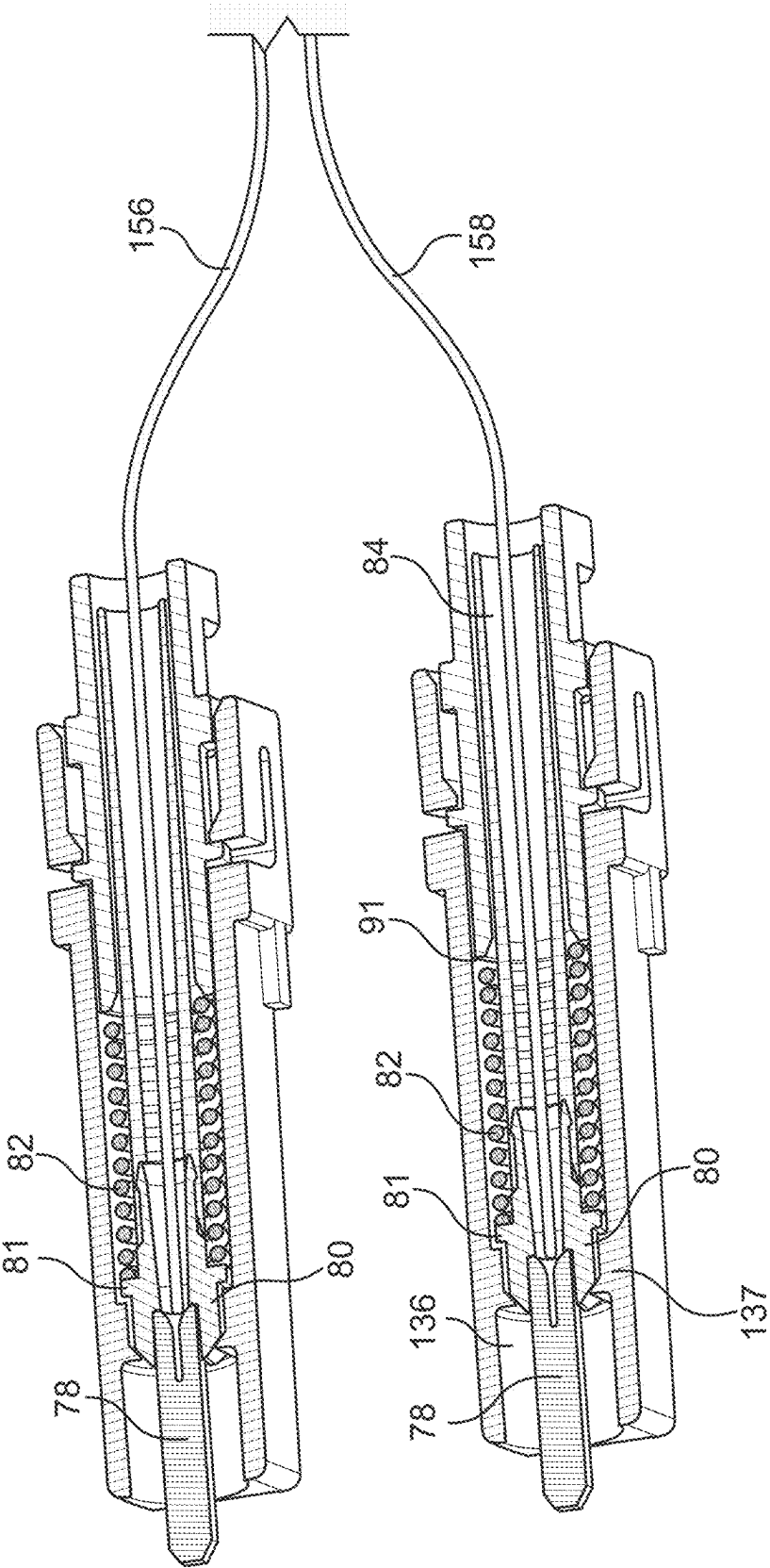


FIG. 16

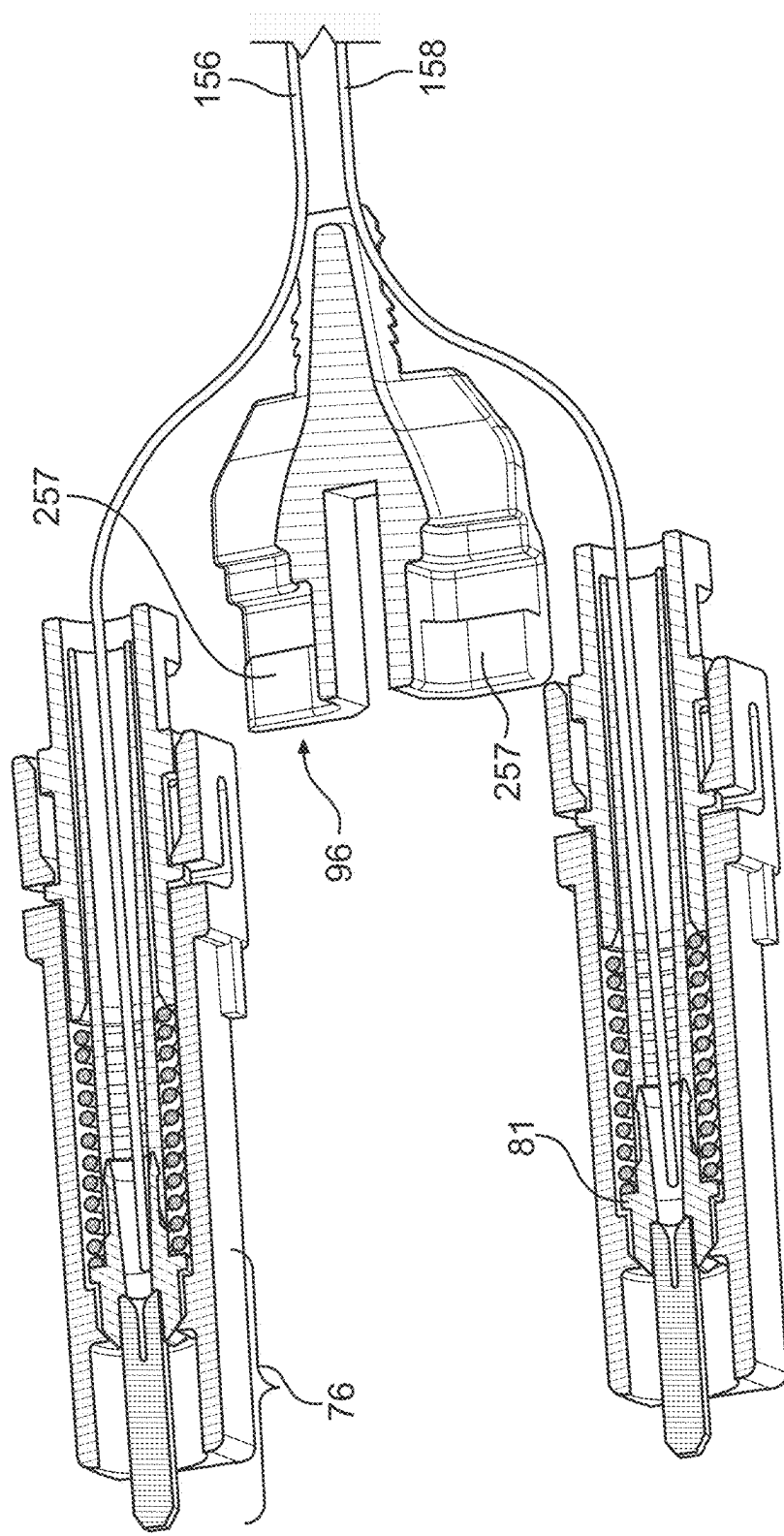


FIG. 17

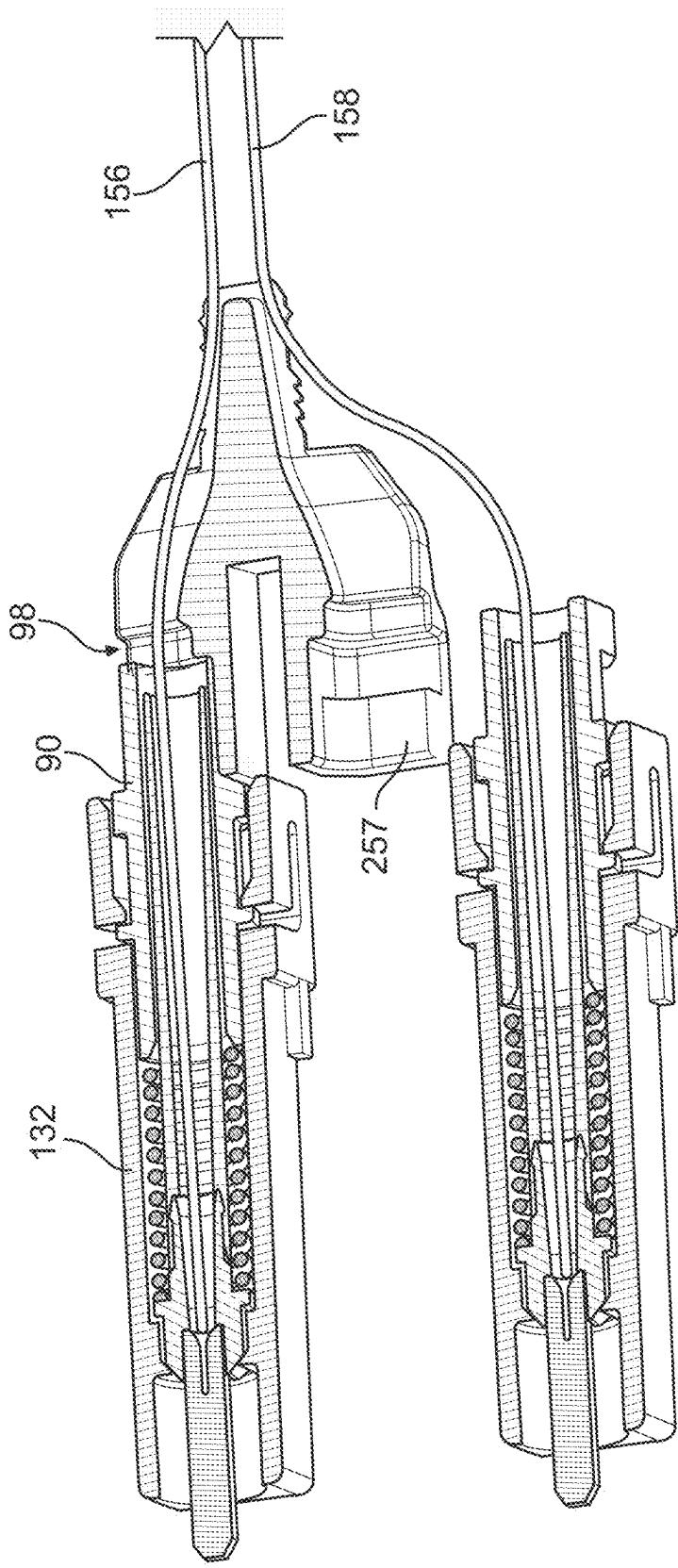
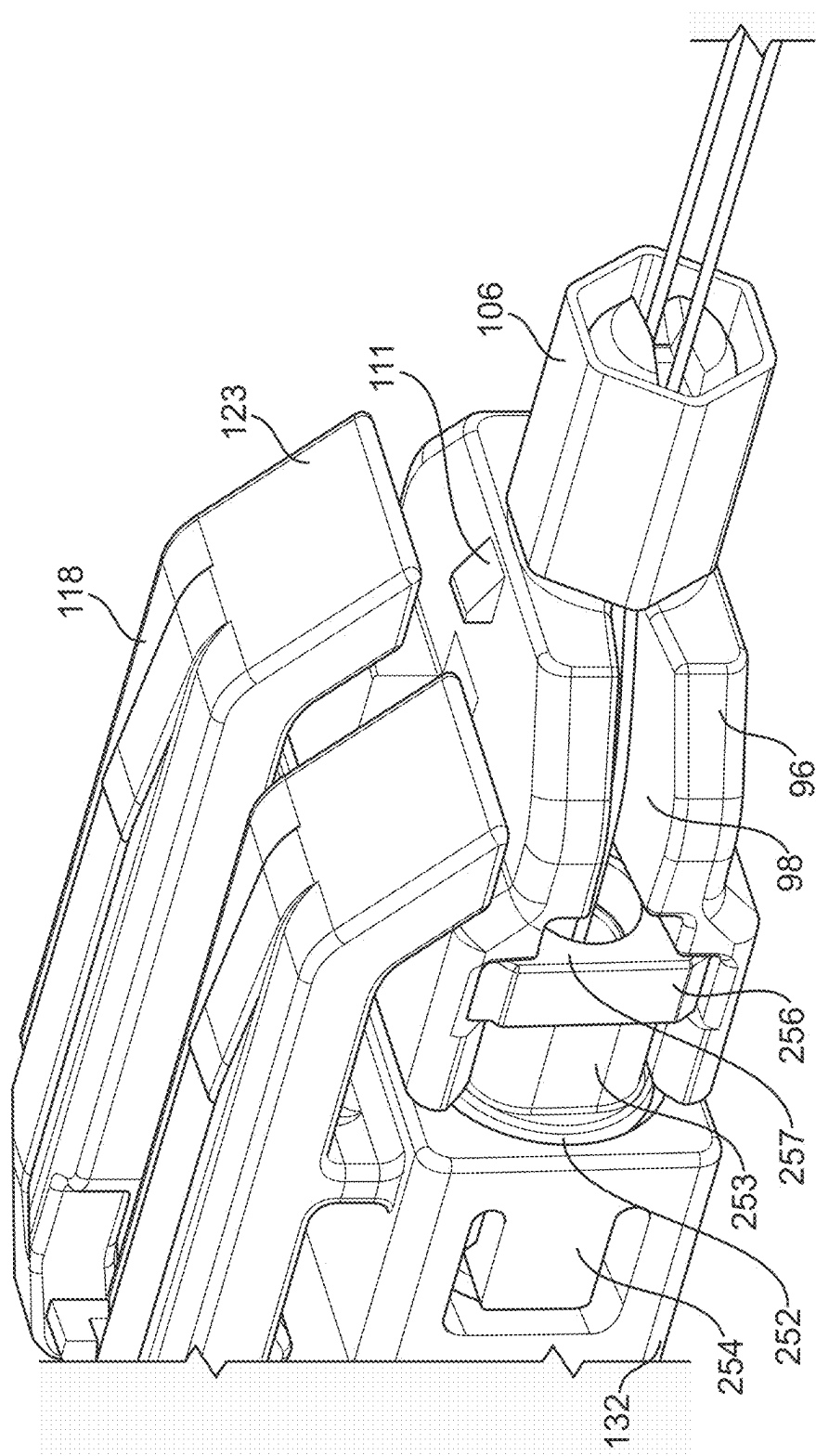


FIG. 18



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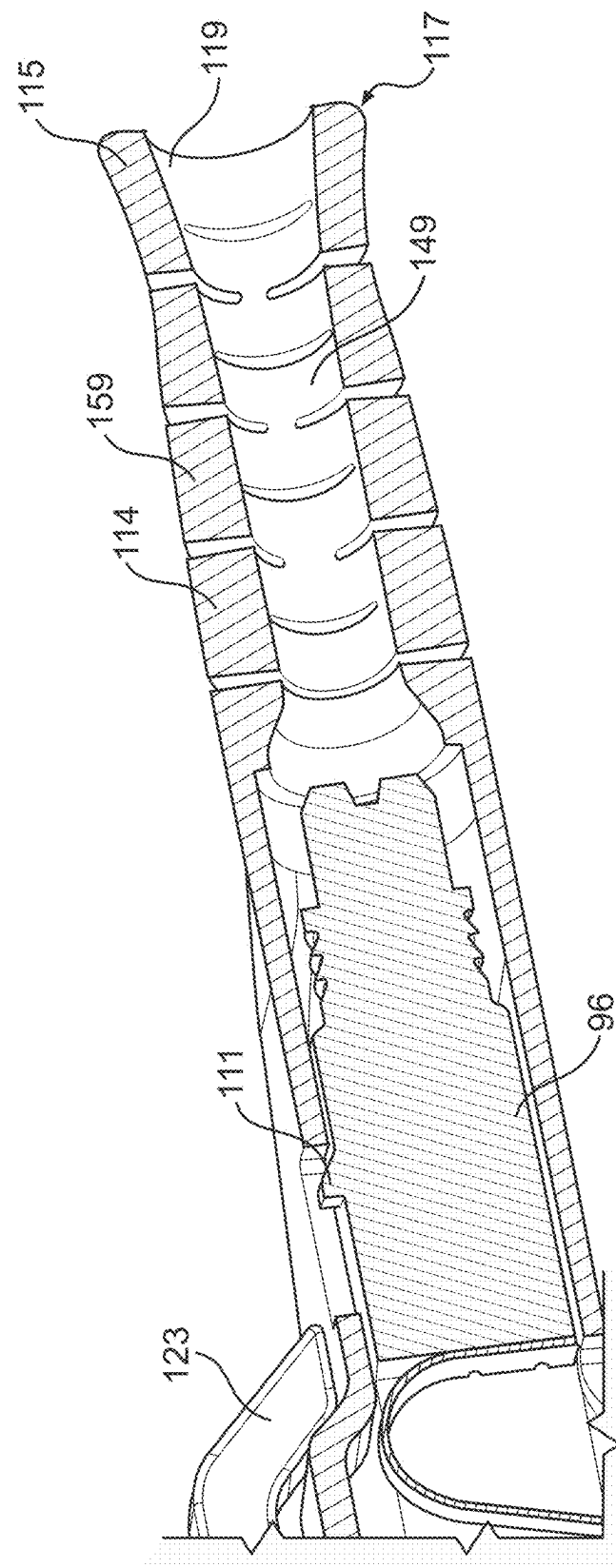


FIG. 20

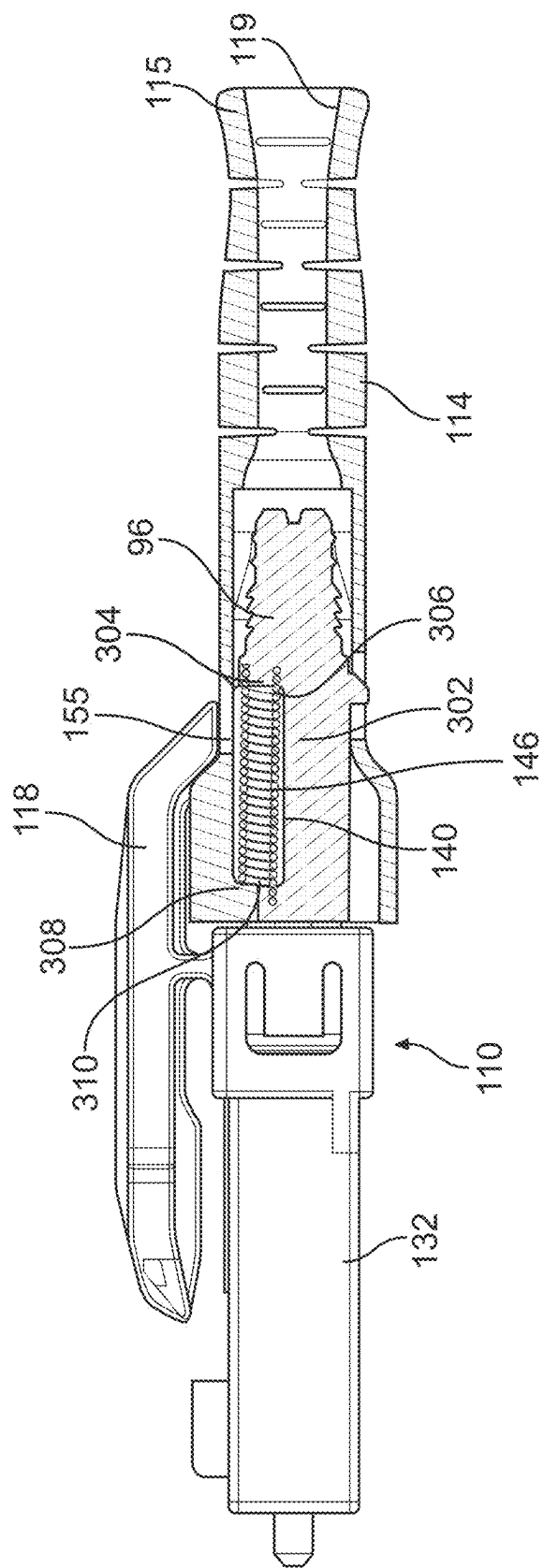


FIG. 21

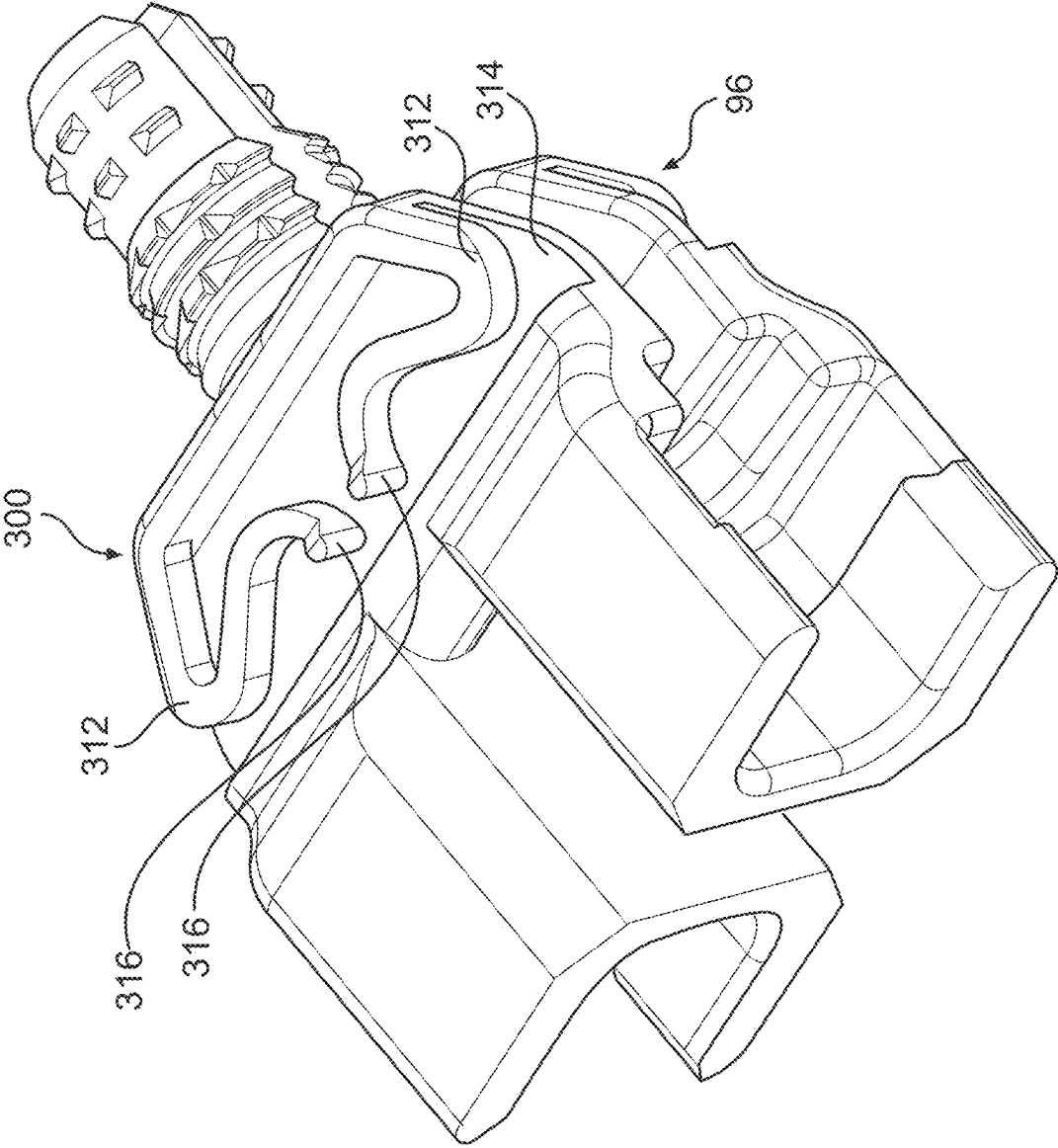


FIG. 22

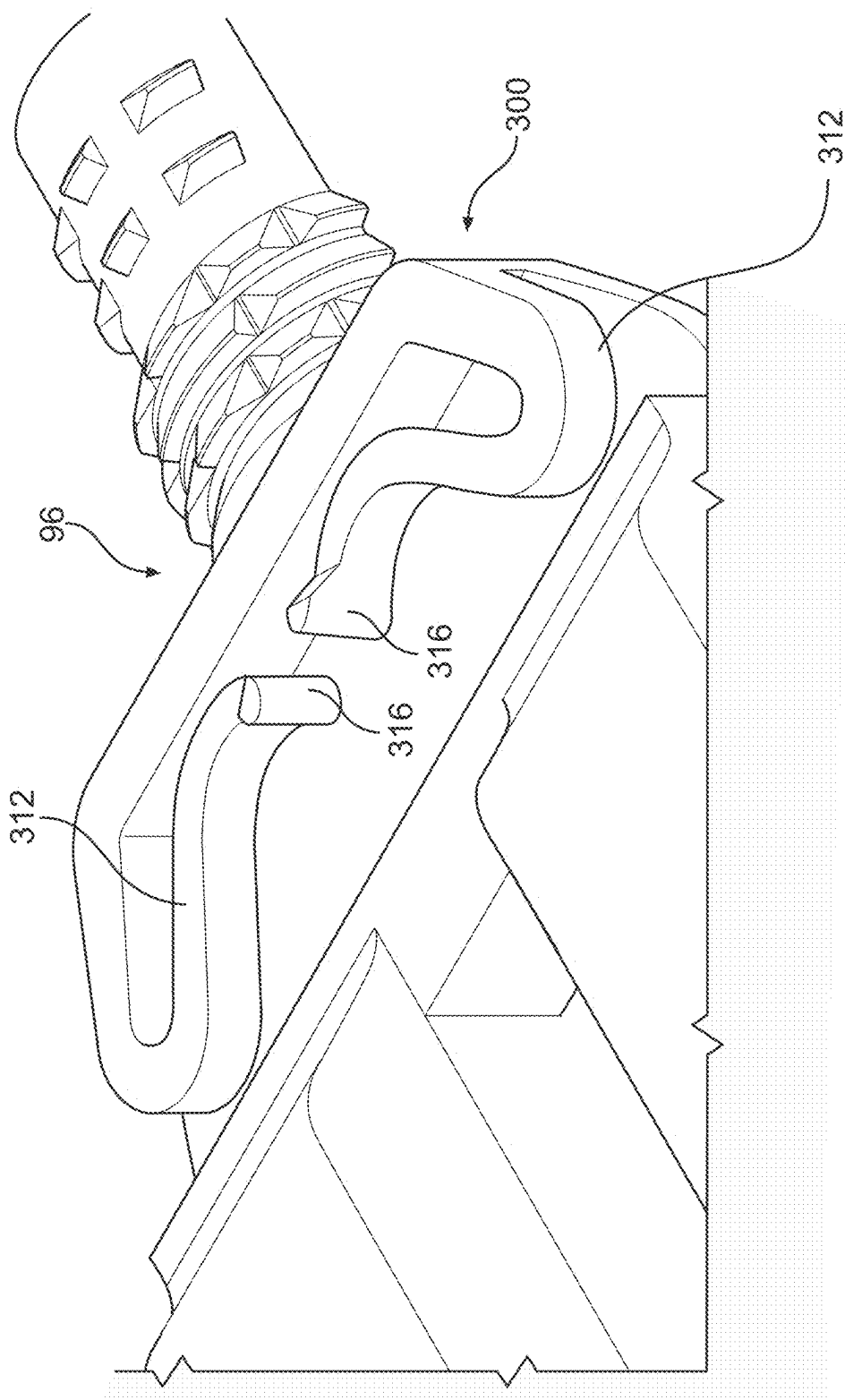


FIG. 23

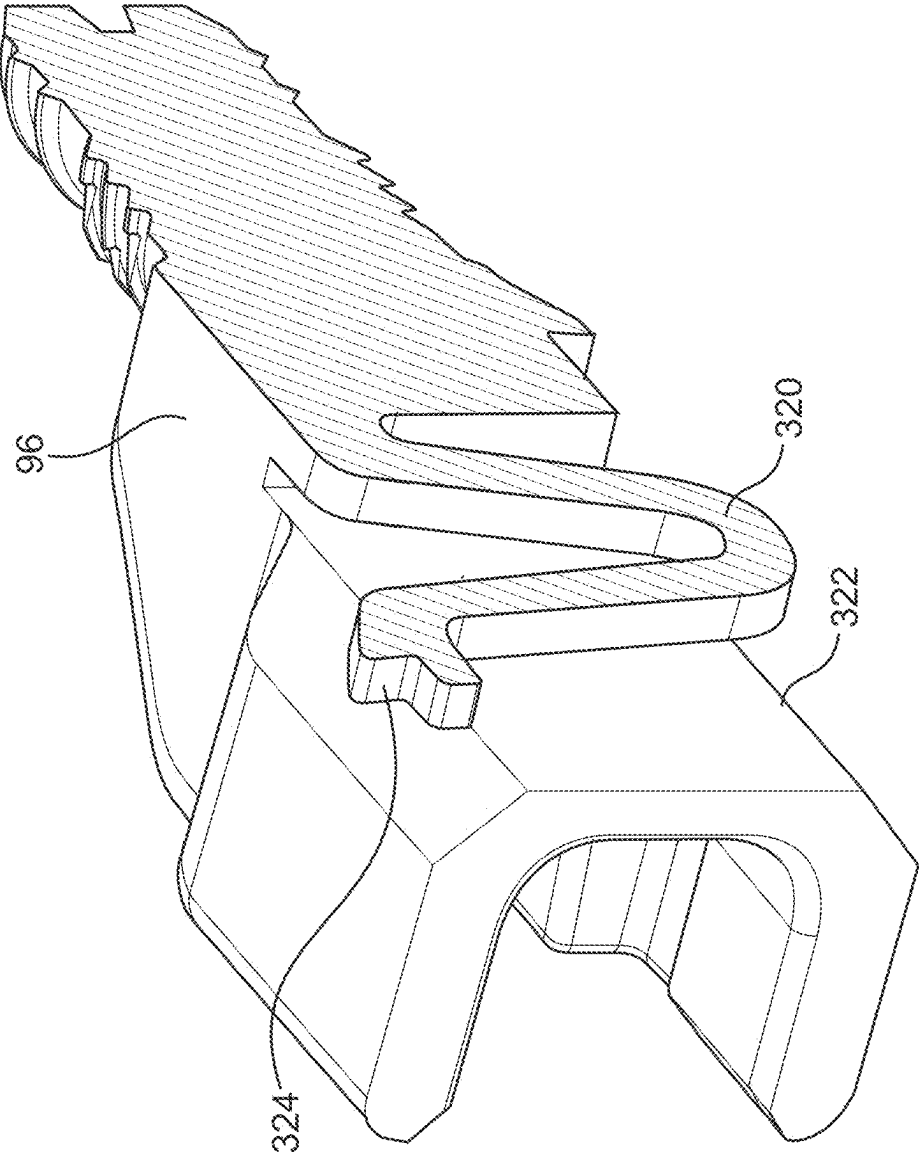


FIG. 24

FIG. 25

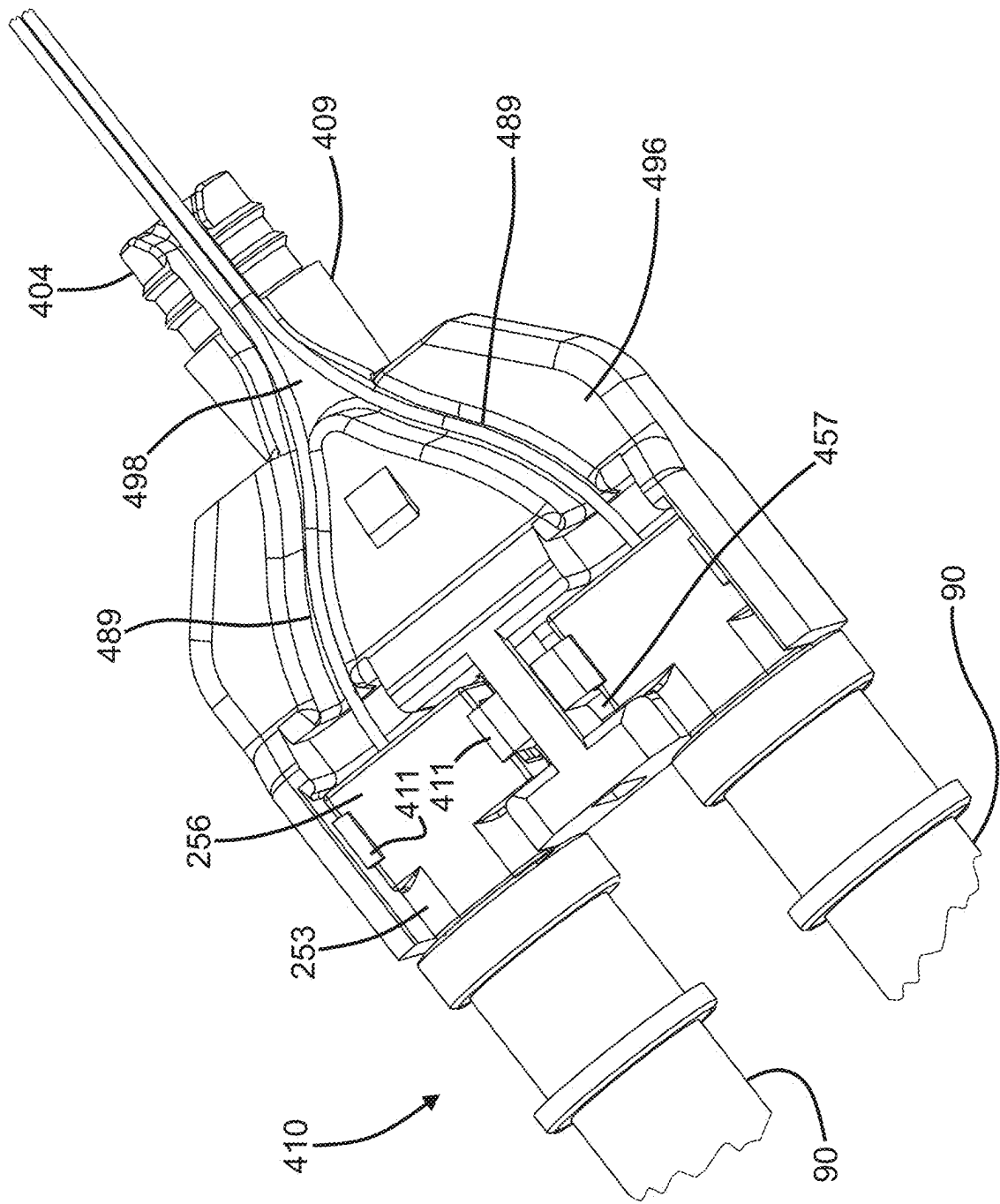
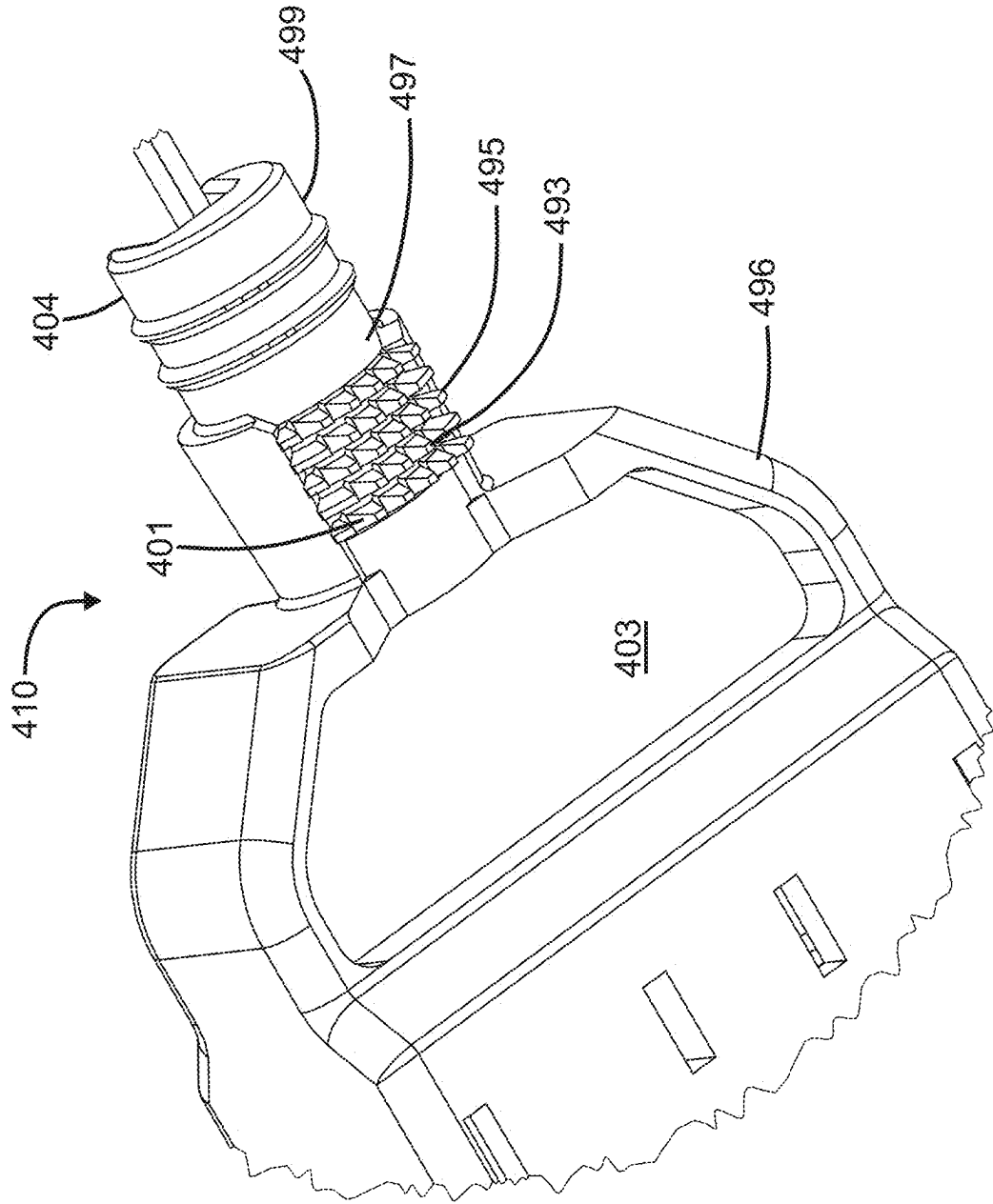


FIG. 26



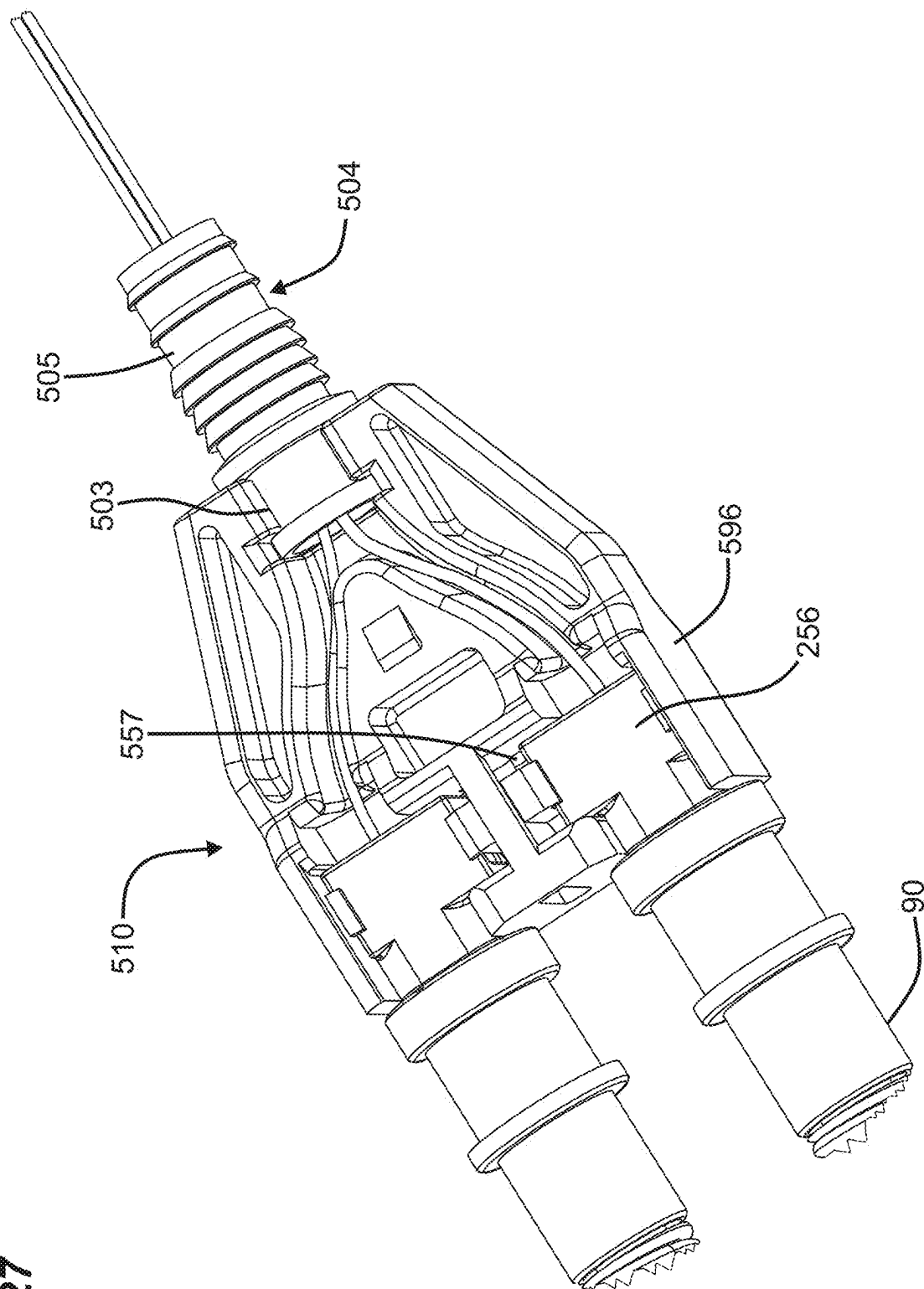
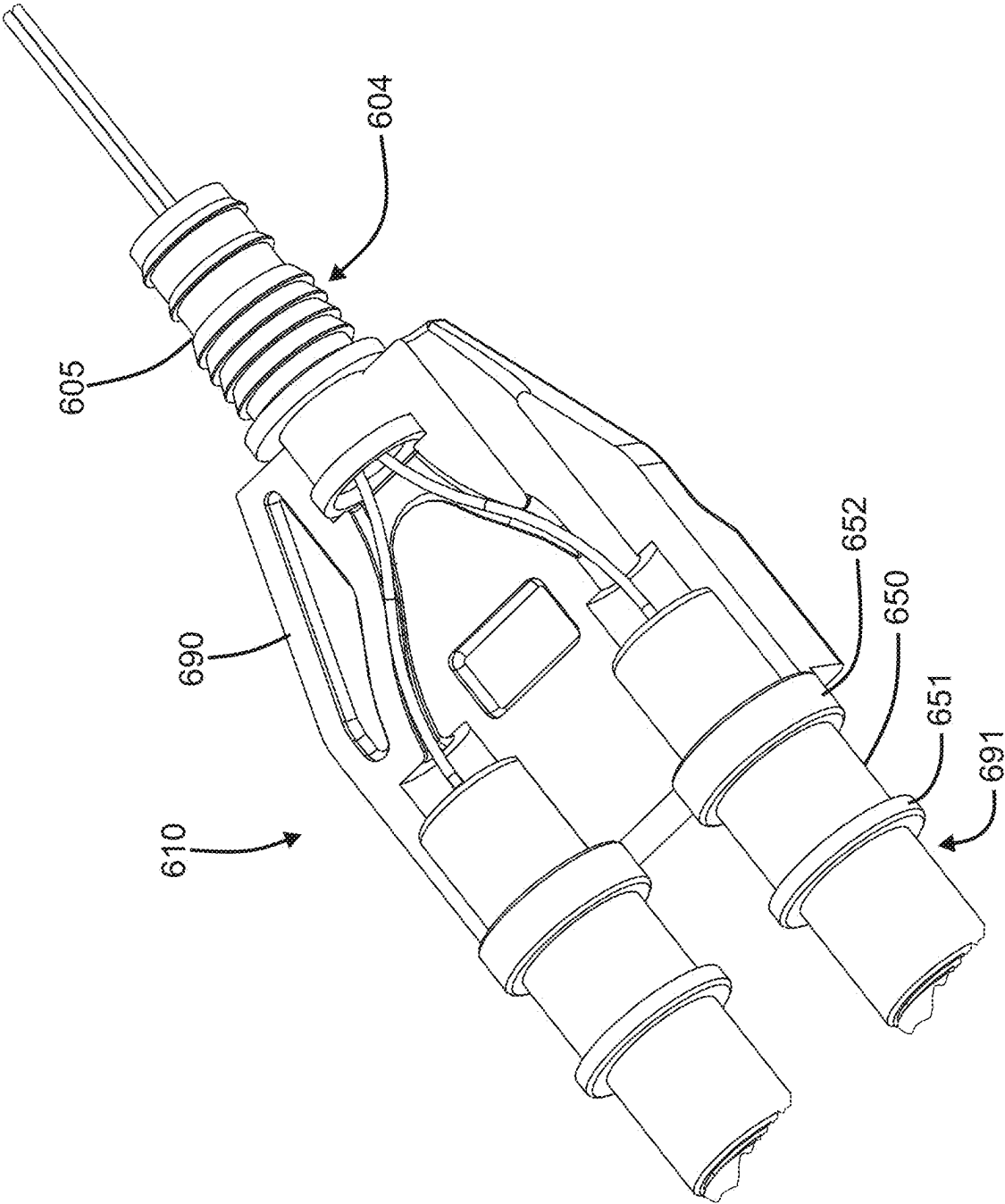


FIG. 27

FIG. 28



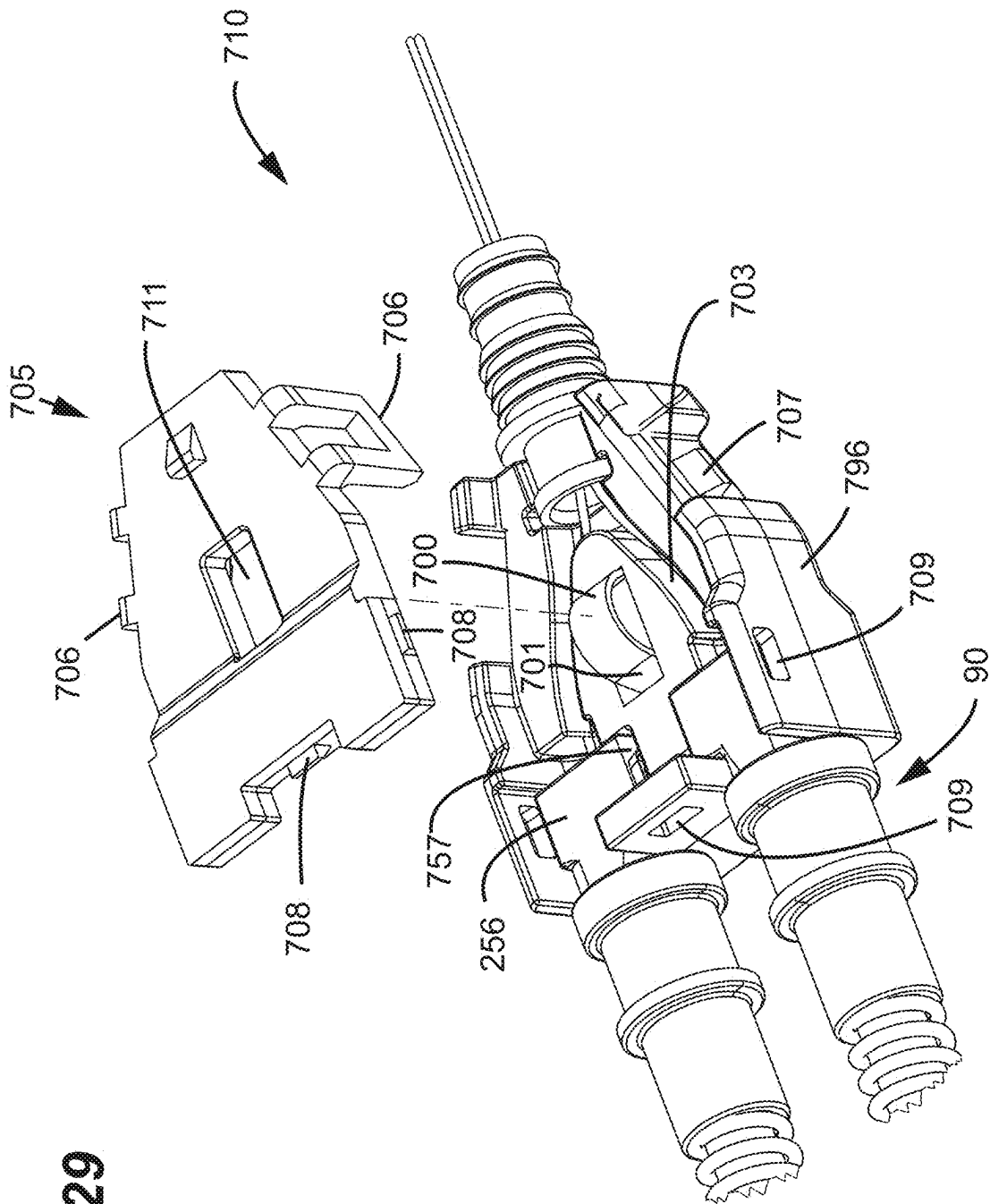


FIG. 29

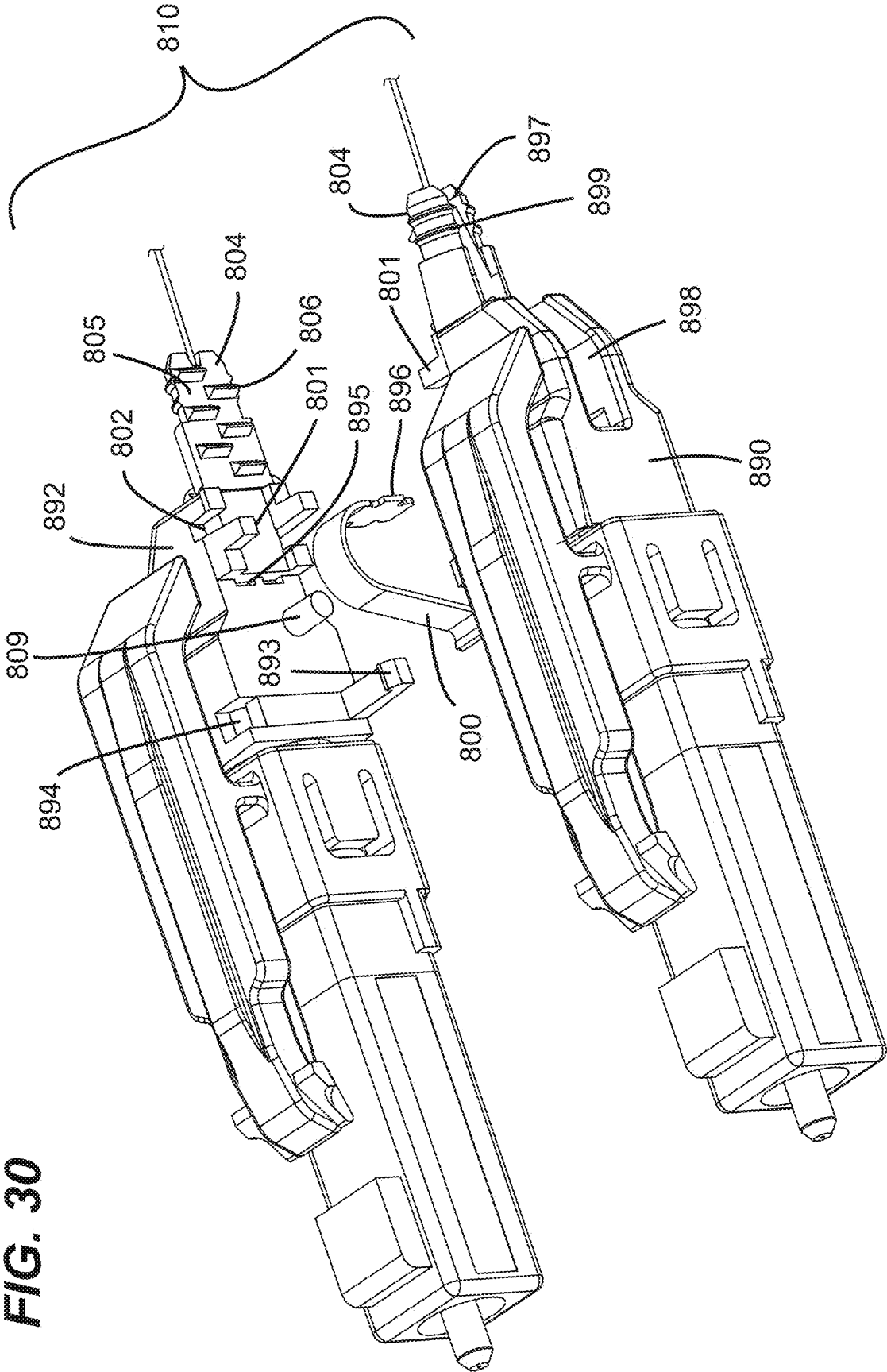


FIG. 31

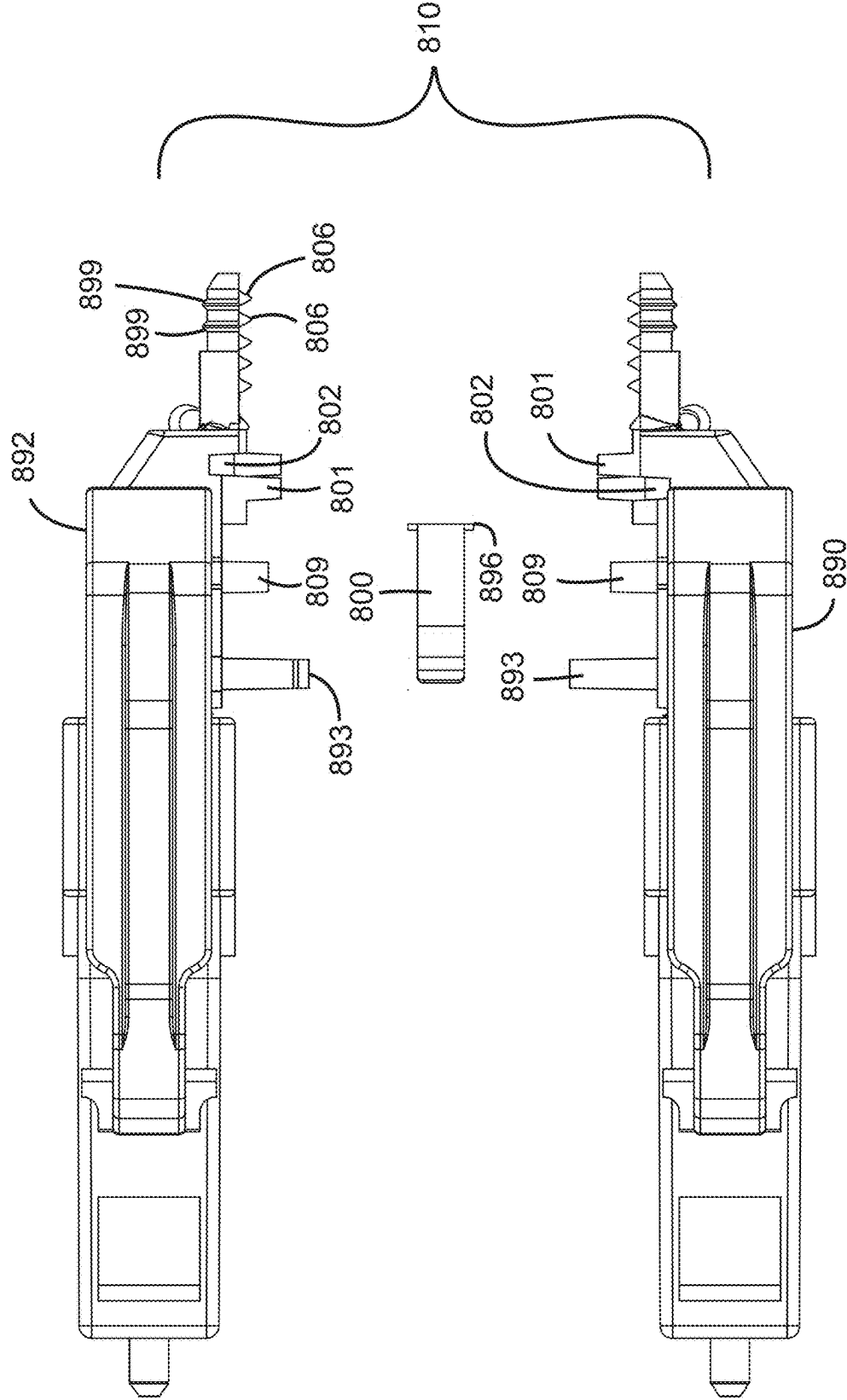
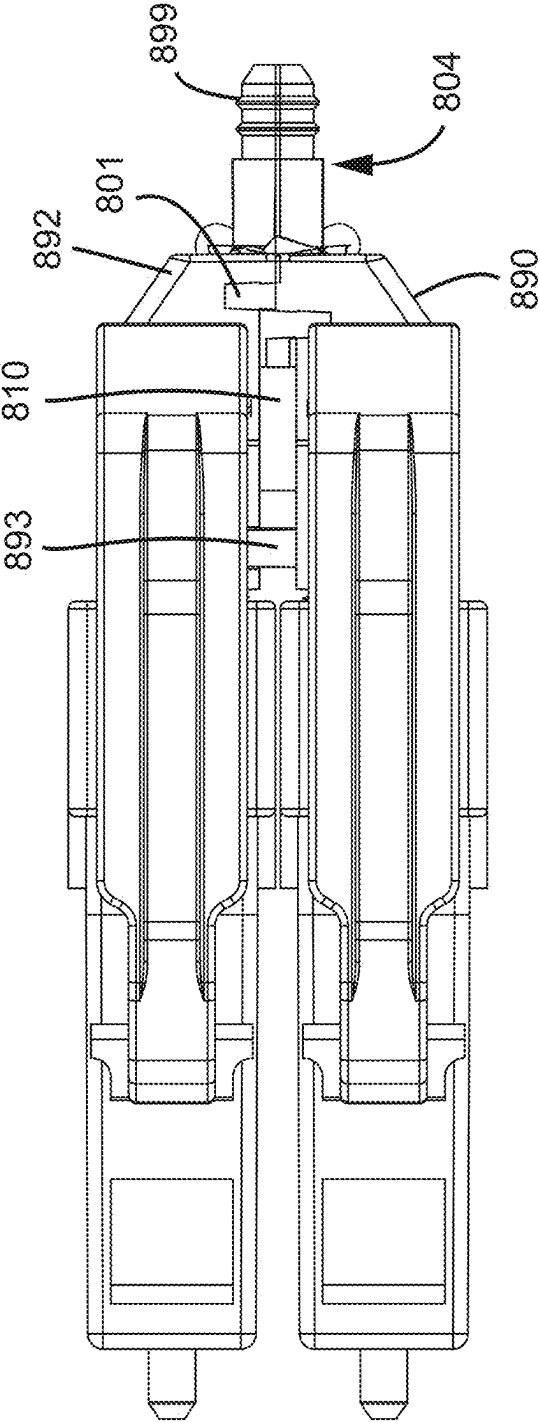


FIG. 32



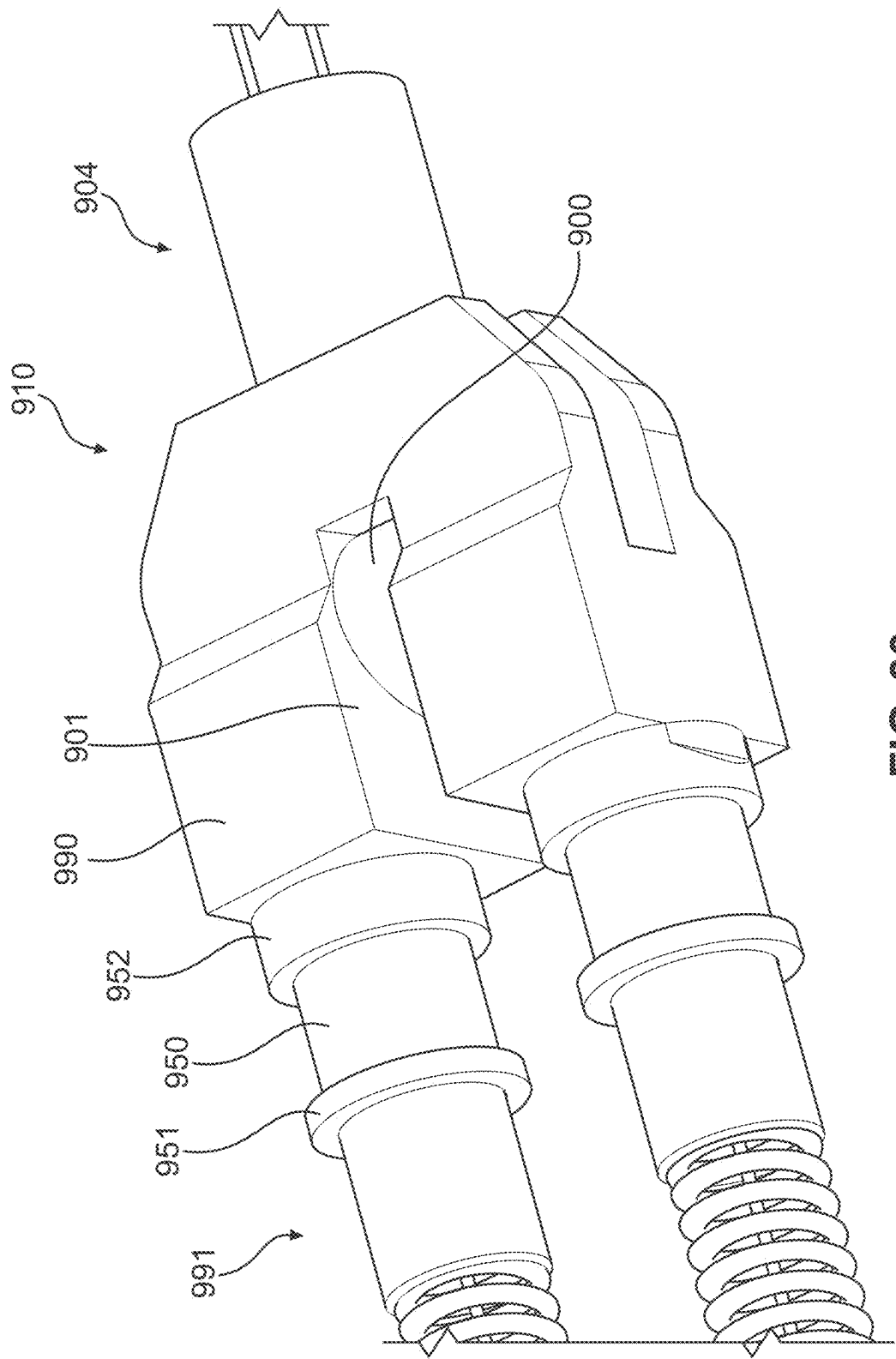


FIG. 33

TELECOMMUNICATIONS CONNECTOR WITH LATCH RELEASE MECHANISM

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is a Continuation of U.S. patent application Ser. No. 18/000,260, filed on Nov. 29, 2022; which is a National Stage Application of PCT/US2021/034612, filed on May 27, 2021; which claims the benefit of U.S. Patent Application Ser. No. 63/032,170, filed on May 29, 2020, the disclosures of which are incorporated herein by reference in their entireties.

TECHNICAL FIELD

[0002] The present disclosure relates generally to telecommunications systems. More particularly, the present disclosure relates to optical fiber communications systems and fiber optic connectors used in optical fiber communications systems.

BACKGROUND

[0003] Fiber optic communications systems are becoming prevalent in part because service providers want to deliver high bandwidth communication capabilities (e.g., data and voice) to customers. Fiber optic communications systems employ a network of fiber optic cables to transmit large volumes of data and voice signals over relatively long distances. Optical fiber connectors are an important part of most fiber optic communication systems. Fiber optic connectors allow two optical fibers to be quickly optically connected without requiring a splice. Fiber optic connectors can be used to optically interconnect two lengths of optical fiber. Fiber optic connectors can also be used to interconnect lengths of optical fiber to passive and active equipment.

[0004] A typical fiber optic connector includes a ferrule assembly supported at a front end of a connector housing. A spring is used to bias the ferrule assembly in a forward direction relative to the connector housing. The ferrule functions to support an end portion of at least one optical fiber (in the case of a multi-fiber ferrule, the ends of multiple fibers are supported). The ferrule has a front end face at which a polished end of the optical fiber is located. When two fiber optic connectors are interconnected, the front end faces of the ferrules abut one another and the ferrules are forced rearward relative to their respective connector housings against the bias of their respective springs. With the fiber optic connectors connected, their respective optical fibers are coaxially aligned such that the end faces of the optical fibers directly oppose one another. In this way, an optical signal can be transmitted from optical fiber to optical fiber through the aligned end faces of the optical fibers. For many fiber optic connector styles, alignment between two fiber optic connectors is provided through the use of an intermediate fiber optic adapter.

[0005] A fiber optic connector is often secured to the end of a corresponding fiber optic cable by anchoring strength members of the cable to the connector housing of the connector. Anchoring is typically accomplished through the use of conventional techniques such as crimps or adhesives. Anchoring the strength members of the cable to the connector housing is advantageous because it allows tensile load applied to the cable to be transferred from the strength members of the cable directly to the connector housing. In

this way, the tensile load is not transferred to the ferrule assembly of the fiber optic connector. If the tensile load were to be applied to the ferrule assembly, such tensile load could cause the ferrule assembly to be pulled and result in optical disconnection between the connector and its corresponding mated connector.

[0006] An example of a prior art LC style fiber optic connector **20** is illustrated in FIGS. **1** and **2**. As shown in FIGS. **1** and **2**, the conventional LC connector **20** includes a connector housing **22** defining a front housing portion **24** and a rear housing portion **26**. The LC connector **20** includes a ferrule assembly **28** defined by a ferrule **30**, a hub **32**, and a spring **34**. A rear end **36** of the ferrule **30** is secured within the ferrule hub **32**. When the LC connector **20** is assembled, the ferrule hub **32** and the spring **34** are captured between the front housing portion **24** and the rear housing portion **26** of the connector housing **22** and a front end **38** of the ferrule **30** projects outwardly beyond a front end **40** of the connector housing **22**. The spring **34** is configured to bias the ferrule **30** in a forward direction relative to the connector housing **22**. According to some versions, the front housing portion **24** may be formed from a molded plastic.

[0007] In the depicted example of the prior art LC connector **20**, a strain relief boot **56** is slid over a rear end **58** of the rear housing portion **26** and snaps over a boot flange **60** to retain the boot **56** with respect to the connector housing **22**. The rear housing portion **26** may define a crimp region **62** for crimping a fiber optic cable's strength layer to the rear housing portion **26**, normally with the use of a crimp sleeve (not shown). An exterior surface **64** of the rear housing portion **26** defining the crimp region **62** can be textured (e.g., knurled, ridged, provided with small projections, etc.) to assist in retaining the crimp on the housing **22**.

[0008] Still referring to FIGS. **1** and **2**, the front housing portion **24** defines a latch **42** extending from a top wall **44** of the front housing portion **24** toward the rear end **46**, the latch **42** extending at an acute angle with respect to the top wall **44** of the front housing portion **24**. The rear housing portion **26** also includes a latch trigger **48** that extends from the rear end **46** of the front housing portion **24** toward the front end **40**. The latch trigger **48** also extends at an acute angle with respect to the top wall **44**. The latch trigger **48** is configured to come into contact with the latch **42** for flexibly moving the latch **42** downwardly.

[0009] As is known in the art, when the fiber optic connector **20** is placed in an LC format adapter **100**, an example of which is shown in FIGS. **3-5**, for optically coupling light from two optical fibers together, the latch **42** functions to lock the fiber optic connector **20** in place within the adapter **100**. The fiber optic connector **20** may be removed from the adapter **100** by depressing the latch trigger **48**, which causes the latch **42** to be pressed in a downward direction, freeing catch portions **52** of the latch **42** from the fiber optic adapter **100**.

[0010] In dense panel environments, where a large number of connectors, such as connector **20**, are placed in a side-by-side arrangement, access becomes a big issue. Since the latch triggers **48** have to be accessed and pushed down for unlatching the latches **42** from the adapters **100**, removal of connectors can become difficult and cumbersome. In dense panel designs, even though it is normally fairly easy to insert and latch connectors such as connectors **20** by pushing from an end of the connector's boot, removal is difficult because the latches are far forward on the connectors.

[0011] Improvements in connector design are desired to facilitate use in high density environments.

SUMMARY

[0012] One aspect of the present disclosure relates to a telecommunications connector (e.g., a fiber optic connector) having features that facilitate connector latching and unlatching.

[0013] Another aspect of the disclosure relates to a duplex telecommunications connector design, where front housings of two connector portions that are positioned side-by-side can be rotated about their longitudinal axes to change the polarity of the two connector portions.

[0014] According to one aspect, the disclosure is directed to a telecommunications connector, for example, a fiber optic connector, that includes at least one connector portion including a front housing portion coupled to a rear housing portion. The front housing portion is rotatable relative to the rear housing portion about a longitudinal axis defined by the front housing portion, the front housing portion including a ferrule and a latch. The latch defines a front end and a rear end, and is pivotable relative to the rest of the front housing portion at a connection portion. According to one embodiment, the connection portion is an intermediate connection portion positioned between the front end and the rear end of the latch.

[0015] The telecommunications connector also includes a yoke coupled to the rear housing portion of the at least one connector portion, wherein the rear housing portion is rotationally and axially fixed with respect to the yoke, and a boot mounted to the yoke, the boot movable longitudinally relative to the yoke, wherein the boot is configured to engage the rear end of the latch and cause the front end of the latch to pivot toward the ferrule of the front housing portion as the boot is moved away from the yoke.

[0016] According to another aspect of the disclosure, the telecommunications connector is a duplex fiber optic connector that includes two connector portions provided in a side-by-side relationship, each connector portion including a front housing portion coupled to a rear housing portion, wherein each front housing portion is rotatable relative to the rear housing portion about a longitudinal axis defined by the front housing portion to change the polarity of the two connector portions. Each front housing portion includes a ferrule and a latch, wherein the latch defines a front end and a rear end, and is pivotable relative to the rest of the front housing portion at a connection portion. A yoke is coupled to the rear housing portion of each of the connector portions, wherein the rear housing portions are rotationally and axially fixed with respect to the yoke. A boot is mounted to the yoke, the boot movable longitudinally relative to the yoke, wherein the boot is configured to engage the rear ends of each of the latches and cause the front ends of each of the latches to pivot toward the ferrules of the front housing portions as the boot is moved away from the yoke.

[0017] According to another aspect of the disclosure, a telecommunications connector comprises at least one connector portion including a front housing portion coupled to a rear housing portion, wherein the front housing portion is rotatable relative to the rear housing portion about a longitudinal axis defined by the front housing portion, the front housing portion including a ferrule and a latch, wherein the latch defines a front end and a rear end, and is pivotable relative to the rest of the front housing portion at a connection

portion, a yoke coupled to the rear housing portion of the at least one connector portion, wherein the rear housing portion is rotationally and axially fixed with respect to the yoke, and a boot mounted to the yoke, the boot movable longitudinally relative to the yoke, wherein the boot is configured to engage at least a portion of the latch and cause the latch to pivot relative to the rest of the front housing portion as the boot is moved relative to the yoke.

[0018] According to yet another aspect of the disclosure, a telecommunications connector comprises a front housing portion coupled to a rear housing portion and a crimp region, wherein the front housing portion includes a ferrule and a latch, wherein the latch defines a front end and a rear end, and is pivotable relative to the rest of the front housing portion at a connection portion, the crimp region configured for crimping at least a portion of a cable terminated by the ferrule, wherein the crimp region defines a plurality of spaced apart crimp rings stacked longitudinally, each crimp ring defining discretely spaced scalloped portions that are positioned peripherally around the rings, wherein the scalloped portions between adjacent rings are provided in a staggered position in a longitudinal direction defined by the telecommunications connector.

[0019] According to yet another embodiment of the disclosure, a telecommunications connector includes at least one connector portion including a front housing portion coupled to a rear housing portion, wherein the front housing portion is rotatable relative to the rear housing portion about a longitudinal axis defined by the front housing portion, the front housing portion including a ferrule and a latch, wherein the latch defines a front end and a rear end, and is pivotable relative to the rest of the front housing portion at a connection portion, and a boot mounted on the telecommunications connector, the boot movable longitudinally relative to the rear housing portion, wherein the boot is configured to engage at least a portion of the latch of the front housing portion and cause the latch to pivot relative to the rest of the front housing portion as the boot is moved relative to the rear housing portion.

[0020] A variety of additional aspects will be set forth in the description that follows. The aspects relate to individual features and to combinations of features. It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the broad inventive concepts upon which the embodiments disclosed herein are based.

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] FIG. 1 is a top view of a conventional LC style fiber optic connector.

[0022] FIG. 2 is a cross-sectional view that longitudinally bisects the fiber optic connector of FIG. 1.

[0023] FIGS. 3-4 are two perspective views of a conventional duplex adapter for mating with the fiber optic connector of the present disclosure.

[0024] FIG. 5 is a cross-sectional view of the duplex adapter of FIGS. 3-4.

[0025] FIG. 6 is a perspective view of one example fiber optic connector having features that are examples of inventive aspects in accordance with the present disclosure.

[0026] FIG. 7 is a side view of the fiber optic connector of FIG. 6, shown in the neutral position.

[0027] FIG. 8 is a side view of the fiber optic connector of FIG. 6, shown in an unlatched configuration.

[0028] FIG. 9 illustrates the rear end regions of the latches of the fiber optic connector of FIGS. 6-8.

[0029] FIG. 10 illustrates the strain relief boot portion of the fiber optic connector of FIGS. 6-8.

[0030] FIG. 11 is a cross-sectional view illustrating the interaction between the rear end of the latch of the fiber optic connector of FIG. 9 and the boot of FIG. 10.

[0031] FIG. 12 illustrates the front and rear housings of the connector portions of the fiber optic connector of FIGS. 6-8 separated from the yoke and the boot of the fiber optic connector.

[0032] FIG. 13 is a perspective view of the fiber optic connector of FIGS. 6-8 with a portion of the front housing broken away to illustrate the internal features thereof.

[0033] FIG. 14 illustrates the fiber optic connector of FIG. 13 with one of the front housings of a connector portion being rotated to reverse the polarity of that connector portion.

[0034] FIG. 15 is a rear perspective view of the yoke of the fiber optic connector of FIGS. 6-8 shown in isolation.

[0035] FIG. 16 is a partial assembly view of the fiber optic connector of FIGS. 6-8 illustrating cross-sectional views bisecting the front and rear housings of the connector portions.

[0036] FIG. 17 illustrates another partial assembly view of the fiber optic connector of FIG. 16 with a cross-sectional view of the yoke added in.

[0037] FIG. 18 illustrates the partial assembly view of the fiber optic connector of FIG. 17 with one of the connector portions coupled to the yoke.

[0038] FIG. 19 illustrates the connector portions coupled to the yoke of the fiber optic connector of FIGS. 6-8 with a crimp sleeve placed on the crimp region of the yoke.

[0039] FIG. 20 illustrates a cross-sectional view of a rear end of the strain relief boot to illustrate the external flared portion and the enlarged inner radius thereof.

[0040] FIG. 21 illustrates one example embodiment of a biased boot-return feature that can be implemented on the fiber optic connector for returning the boot toward the forward position of FIG. 7.

[0041] FIG. 22 illustrates another example embodiment of a biased boot-return feature that can be implemented on the fiber optic connector for returning the boot toward the forward position of FIG. 7.

[0042] FIG. 23 illustrates the boot-return feature of FIG. 22 in a flexed position.

[0043] FIG. 24 illustrates yet another example embodiment of a biased boot-return feature that can be implemented on the fiber optic connector for returning the boot toward the forward position of FIG. 7.

[0044] FIG. 25 illustrates a top perspective, cross-sectional view of a portion of another embodiment of a fiber optic connector having features that are examples of inventive aspects in accordance with the present disclosure.

[0045] FIG. 26 illustrates a bottom perspective, cross-sectional view of a portion of the fiber optic connector of FIG. 25.

[0046] FIG. 27 illustrates a top perspective, cross-sectional view of a portion of another embodiment of a fiber optic connector having features that are examples of inventive aspects in accordance with the present disclosure.

[0047] FIG. 28 illustrates a top perspective, cross-sectional view of a portion of yet another embodiment of a fiber optic connector having features that are examples of inventive aspects in accordance with the present disclosure.

[0048] FIG. 29 illustrates a top perspective view of a portion of yet another embodiment of a fiber optic connector having features that are examples of inventive aspects in accordance with the present disclosure.

[0049] FIG. 30 illustrates a top perspective exploded view of a portion of yet another embodiment of a fiber optic connector having features that are examples of inventive aspects in accordance with the present disclosure.

[0050] FIG. 31 illustrates a top view of the fiber optic connector of FIG. 30.

[0051] FIG. 32 illustrates a top view of the connector of FIGS. 30 and 31 in an assembled configuration.

[0052] FIG. 33 illustrates a top perspective view of a portion of yet another embodiment of a fiber optic connector having features that are examples of inventive aspects in accordance with the present disclosure, the fiber optic connector having features similar to the connector of FIG. 28.

DETAILED DESCRIPTION

[0053] Referring now to FIGS. 6-24, an example embodiment of a connector 110 having features that are examples of inventive aspects in accordance with the present disclosure is illustrated. The illustrated connector 110 is a fiber optic connector that is used for aligning optical fibers. However, it should be noted that the inventive features discussed herein are fully applicable to other types of connectors that are not necessarily fiber optic connectors.

[0054] The connector 110 may be considered a duplex LC format connector, transitioning optical fibers from a single optical cable to two front fiber optic connector portions 112 as will be discussed in further detail below. In the illustrated embodiment, each connector portion 112 may define an LC profile, meaning that the connector portions 112 can mate to other LC type connectors via an LC adapter, such as the adapter 100 shown in FIGS. 3-5, as noted above. As such, it should be noted that certain structures and functionality of those structures may be similar to the simplex LC connector 20 discussed above with respect to FIGS. 1-2.

[0055] Still referring now to FIGS. 6-24, the connector portions 112 of the illustrated connector 110 each houses a ferrule 78 for terminating the optical fiber of a fiber optic cable. As noted above, connector 110 may also be referred to as duplex connector assembly or simply a duplex connector. As noted above, connector 110 is mateable via the adapter 100 shown in FIGS. 3-5. Adapter 100 mates two of the connectors 110 together or mates another LC format connector to connector 110 for fiber optic signal transmission.

[0056] In the depicted embodiment of the connector 110, the two connector portions 112 are arranged side-by-side and in a parallel position, as illustrated in FIG. 6. Each connector portion 112 is formed from a front housing 132 that is coupled to and rotatable relative to a rear housing 90. A ferrule assembly 76 is positioned within the front housing 132. Ferrule assembly 76 includes the ferrule 78, a hub 80 which holds the ferrule 78, and a spring 82 which is positioned between a rear shoulder 81 of the ferrule hub 80 and a front end 91 of the rear housing 90. The spring 82 is configured to bias hub 80 and ferrule 78 forwardly.

[0057] As shown in the internal details of the connector 110 in FIGS. 13 and 14, an internal tube 84 may be provided extending from the ferrule hub 80. Tube 84 may be configured to prevent epoxy from interfering with the movement of the ferrule 78, the hub 80, and the spring 82.

[0058] As will be discussed in further detail below, both of the rear housings 90 of the two connector portions 112 that are positioned in a side-by-side arrangement are received by a yoke 96 through side slots 98 defined by the yoke 96. The yoke is illustrated in isolation in FIG. 15.

[0059] The rear housings 90 are rotationally fixed with respect to the yoke 96 once laterally received therein. As will be discussed in further detail below, it is the front housings 132 of the connector portions 112 that are rotatable relative to the rear housings 90 and the yoke 96 to change the polarity of the overall connector 110, as illustrated in FIGS. 13-14.

[0060] The yoke 96 defines a rear crimp region 104 for allowing crimping portions of a fiber optic cable via a crimp sleeve 106, as shown in FIGS. 15 and 19.

[0061] In an example embodiment, a fiber optic cable may include an outer jacket, strength members, typically in the form of aramid yarns, and two fibers 156, 158 as illustrated in the present disclosure. Each fiber 156, 158 is to be terminated to one of the two connector portions 112 of the connector 110. Each fiber 156, 158 may typically include an outer coating and a bare fiber. Typically, the coating is removed and the bare fiber is inserted into the ferrule 78, and affixed, such as with epoxy. The aramid yarns are crimped to the crimp region 104 of the yoke 96 as will be discussed in further detail below.

[0062] Referring now to FIGS. 7-9, each front housing 132 of the connector portions 112 defines a latch 118 with a front end 124 and a rear end 122. Latch 118 is designed with a certain amount of elastic flexibility to pivot around a connection point 126 during latching and unlatching of latch 118 from an adapter such as adapter 100. Latch 118 is the portion of the connector 110 that is designed to secure connector 110 to adapter 100. The latch 118 functions similar to that of a conventional LC connector as shown in FIGS. 1-2 in coupling the connector 110 to the adapter 100. The latch 118 includes a pair of shoulders 128 at the front end 124 (as shown in FIG. 6) that mate with latching shoulders 208 of adapter 100 to secure the connector 110 to the adapter 100 (as shown in FIGS. 3-5). As noted, in the illustrated embodiment, each connector portion 112 defines an LC profile, meaning that the connector portion 112 can mate with an LC format adapter. The rear end 122 of each latch includes an angled-down portion 123 that is configured to interact with a strain relief boot of the connector 110 as will be discussed in further detail below.

[0063] Illustrated in further detail in FIGS. 13, 14, and 16-18, each front housing 132 includes an inner passage 136 for receiving the ferrule assembly 76. The inner passage 136 includes a hub mount 137 toward the front of the front housing 132 that defines a keyed configuration for receiving the ferrule hub 80 in a predetermined keyed orientation. The exterior surface of the ferrule hub 80 is designed with a plurality of flats 85 for rotationally tuning the ferrule hub 80 and axially inserting the ferrule hub 80 within the hub mount 137 in a certain keyed orientation. The flats 85 defined on the surface of the ferrule hub 80 and mating surfaces of the hub mount 137 rotationally fix the ferrule hub 80 with respect to

the front housing 132. In this manner, when the front housing is rotated, the ferrule hub 80 rotates therewith.

[0064] The hub mount 137 defines a tapered front shoulder 139 for contacting the front of the ferrule hub 80 and limiting further forward movement of the ferrule hub 80. As noted above, when the front housing 132 rotates, the ferrule hub 80 and the epoxied ferrule 78, including the optical fiber 156/158 terminated thereto, rotates with the front housing 132, all relative to the rear housing 90, as shown in FIG. 14.

[0065] The front housings 132 are inserted into, for mating with similar format connectors, inner passages 202 of adapter 100. The adapter 100 includes two side-by-side passages 202 and two ferrule alignment sleeves 204 there-within. Each alignment sleeve 204 is configured to axially align the ferrules 78 for mating two similar connectors.

[0066] The front housings 132 can be unitarily molded with the latches 118 as a one-piece element. In the depicted example, the front housings 132 define an LC profile for mating with ports 210 of adapter 100. While the example connector 110 is depicted as a duplex connector that includes two fiber optic connector portions 112, it is to be appreciated that the inventive features discussed herein are fully applicable to a connector 110 that may include a single connector portion 112.

[0067] As mentioned above, the connector 110 of the present disclosure is designed such that a strain relief boot 114 that is mounted to and slidable with respect to the yoke portion 96 of the connector allows the latching/unlatching functionality of the connector 110. The strain relief boot is illustrated in further detail in FIGS. 10, 11, 13, 14, and 20.

[0068] Referring to FIGS. 10, 13, 14, and 20, the boot 114 defines a throughhole 149. The throughhole 149 forms a front opening 154 for receiving the yoke 96. Boot 114 includes a flexible rear portion 159 to act as strain relief for the optical fibers 156, 158 that pass through the boot 114. The boot 114 is designed to be movable away from connector portions 112, also relative to the yoke 96, in a longitudinal direction (Direction A in FIG. 8) causing pivoting movement of latch 118 about connection point 126 (Direction B in FIG. 8). Such pivoting movement allows for unlatching of connector portions 112 from adapter 100. Boot 114 simultaneously moves both latches 118 to allow for connector 110 to be unlatched from a duplex adapter or adapters 100 with side-by-side ports 210.

[0069] Referring now to FIG. 20, a flared exterior portion 115 at a rear end 117 of the flexible rear portion 159 of the boot 114 acts as a grip portion to facilitate pulling of the boot 114 rearward for unlatching the connector 110. As illustrated in FIG. 20, the rear end of the opening or throughhole 149 of the boot 114 also follows the general shape of the flared exterior portion 115 to provide a flared interior surface 119. The flared, enlarged, interior surface 119 defines an inside radius that limits sharp cable bends when load is applied to the cable, to decrease bending loss.

[0070] Still referring to the boot 114 as shown in FIGS. 7-10, 13, 14, and 20, boot 114 defines a sloped exterior surface 151 adjacent the front end that is configured to interact with the angled portions 123 of the rear ends 122 of the latches 118 during slidable movement of the boot 114. A lifting motion results for the rear ends 122, which results in a downward movement of front ends 124 of latch 118, when boot 114 is pulled longitudinally away from a remainder of connector 110. Compare FIGS. 7 and 8. A user can pull on boot 114 in a longitudinal direction away from the ferrules

78, and remove the connector 110 from the adapter 100, without directly engaging the latches 118.

[0071] As shown in detail in FIGS. 9-11, the boot 114 also includes a pair of side-by-side slots or recesses 153 which receive latch protrusions 113 defined underneath the angled portions 123 of the rear ends 122 of latches 118. The protrusions 113 and the recesses 153 interact as stops during polarity reversal of the connector portions 112 such that the front housings 132 end up in the correct position with respect to the rear housings 90 after rotation. The exiting sides of the recesses 153 include a bevel 109 to facilitate rotation of the latch 118 out of a given recess 153 with little force. The bevels 109 help with starting rotation of the front housings 132 for polarity reversal.

[0072] The boot 114 includes similar slots 153 on an opposing side of the boot 114. Thus, the boot 114 stays stationary during rotation of the front housings 132 for polarity reversal.

[0073] For the latching and unlatching functionality, a tab 111 defined on the yoke 96 (as shown in FIGS. 11 and 15) mates with an aperture 155 defined on the boot 114 to allow longitudinal movement of boot 114 relative to yoke 96. The tab 111, however, is configured to abut a front end 107 of the aperture 155 to act as a stop for the slidable boot 114.

[0074] Now referring generally to FIGS. 16-19, to assemble the connector 110, the fiber optic cable is initially inserted through boot 114 and the crimp sleeve 106. The fibers 156, 158 are inserted through the rear housings 90 and the tubes 84 and affixed to the ferrules 78 of the ferrule assemblies 76. The ferrule assemblies 76 are keyed as discussed above and placed within the front housings 132. As shown in FIGS. 16-18, the rear housings 90 are axially inserted and snapped into the front housings 132 with springs 82 positioned between the ferrule hubs 80 and the rear housings 90. The tubes 84 extend rearward from the ferrule hubs 80 and are partially positioned within both the front housings 132 and the rear housings 90.

[0075] As noted above, the rear housings 90 and the optical fibers 156, 158 are inserted in a lateral direction through the side slots 98 of the yoke 96, as shown in FIGS. 17-19. The side slots 98 of the yoke 96 enable the fibers 156, 158 to come in from the sides of the yoke 96 and allow the connector portions 112 to be epoxied and polished before the cable has to be crimped to the overall connector 110. It should be noted that the boot 114 may define internal ribs that are configured to protrude into the side slots 98 of the yoke 96 for the purposes of constraining the fibers 156, 158 within the side slots 98 and limit deflection of the fibers. As such, if the ferrule(s) 78 are pressed rearward under a bias and the terminated fibers 156, 158 are pushed back inside the connector 110, the fibers 156, 158 are forced back into the jacket of the cable, guided by a predetermined path provided within the yoke 96, to limit deflection or unwanted bending of the fibers.

[0076] Once the connector portions 112 have been placed on the yoke 96 with rear housings 90 having been inserted laterally into the yoke 96, the cable jacket and strength members in the form of aramid yarns may be crimped to the crimp region 104 of the yoke 96 via the crimp sleeve 106, as shown in FIG. 19. As shown, the crimp region 104 is provided as a rear projection 99 that extends from the yoke 96.

[0077] Referring generally to FIGS. 15 and 19, an exterior surface 97 of the rear projection 99 that defines the crimp

region 104 can be textured (e.g., knurled, ridged, provided with small projections, etc.) to assist in retaining the strength members (e.g., aramid yarns) on the yoke 96. In the depicted example, the exterior surface 97 defines a series of parallel rings or ridges 95 extending in a direction from the rear end of the rear projection 99 toward the main body of the yoke 96. As shown, each of the rings 95 define discretely spaced scalloped portions 93 that are positioned peripherally around the rings 95. The scalloped portions 93 define teeth 201 thereinbetween. As illustrated, the scalloped portions 93 (and the teeth 201) are provided in an offset or staggered relationship between adjacent rings 95 as the rings 95 extend front to back. In this manner, when the strength members are pressed underneath the crimp sleeve 106 against the exterior surface 97 of the rear projection 99, the strength members will tend to spread out and follow zig-zagging patterns as they are captured underneath the crimp sleeve 106, leading to increased yarn and jacket retention. The staggered design of the scallops 93 and the teeth 201 provided therebetween force the strength members laterally into the scallops 93 in a zig-zagging pattern, such that the strength members do not simply lay parallel to the rear projection 99 in a front-to-back direction, but make multiple lateral bends over multiple edges, to increase the amount of pull force needed to separate the cable from the connector 110.

[0078] As shown in FIG. 15, the rear projection 99 may include a step down portion 89 with full annular rings 95 that are smaller in diameter for engaging with the jacket of the cable and providing jacket retention.

[0079] Once the cable is crimped to the yoke 96, the boot 114 is pulled over the yoke 96 as the yoke enters the front opening 154 of the boot 114 until the rear ends 122 of the latches 118 are in the slots 153 of boot 114 and the tab 111 of the yoke 96 is within the aperture 155 defined on the boot 114 (as shown in FIGS. 9-11). As noted previously, the tab 111 is configured to abut a front end of the aperture 155 to act as a stop for the boot 114 when the boot 114 is slid rearwardly.

[0080] Now referring specifically to FIGS. 12-14, the polarity reversal functionality of the connector 110 will be discussed in further detail. As noted above, in the illustrated example of the connector 110, the front housing 132 includes the ferrule assembly 76 that includes the ferrule hub 80, which is keyed and rotationally fixed to the front housing 132 such that the ferrule hub 80 and thus, the ferrule 78 rotate with the front housing 132 during polarity reversal.

[0081] To switch polarity of connector portions 112, the front housings 132 are rotated in opposite directions so that the rear ends 122 of the latches 118 are moved from slots 153 on a first side of the boot 114 to slots 153 positioned on the opposite side of the boot 114. During polarity switching, boot 114 remains coupled to the yoke 96.

[0082] During the polarity switch, the front housing 132 slidably rotates over the rear housing 90 to change the polarity of the overall connector 110.

[0083] The rear housing 90 forms a front annular recess or notch 250 that is defined between a front annular ring 251 and a center annular ring 252. As shown in FIG. 12, flexible portions 254 of the sidewalls of the front housing 132 fit within the annular recess 250 when the rear housing 90 is fitted within the front housing 132. In this manner, the front housing 132 is rotatable relative to the rear housing 90, but the front and rear housings are axially fixed with respect to each other.

[0084] The rear housing 90 also forms a rear annular recess or notch 253 positioned between the center annular ring 252 and a rear shoulder 256. Both the rear annular recess 253 and the rear shoulder 256 define a generally square cross-sectional profile.

[0085] The rear housings 90, as noted above, are received by the yoke 96 through the side slots 98 defined by the yoke 96. The side slots 98 define C-shaped front cut-outs 257 at the front of the yoke 96. The portion of the rear housing 90 that includes the square rear annular recess 253 and the rear shoulder 256 are laterally and slidably positioned within the C-shaped cut-outs 257.

[0086] The internal configuration of the side slots 98, including the C-shaped cut-outs 257, of the yoke 96 and the rear annular recess 253 defined by the rear housing 90, including the rear shoulder 256, intermate to rotationally fix the rear housing 90 with respect to the yoke 96. Thus, when the front housing 132 is rotated with respect to the rear housing 90, the rear housing 90 and the yoke 96 (and the boot 114, which is rotationally fixed with respect to the yoke 96) stay stationary.

[0087] As noted above, the internal tube 84, which is configured to prevent excess epoxy from leaving the ferrule hub portion 80, is positioned within both the front housing 132 and the rear housing 90 and extends rearward from the ferrule hub 80. Since the ferrule hub 80 is rotationally fixed with respect to the front housing 132 and rotates with the front housing 132, the tube 84 may also rotate with the ferrule hub 80 depending upon the strength of the relative frictional engagement between the tube 84 and the ferrule hub 80.

[0088] Now referring to FIGS. 7-8 and 21-24, even though the latch 118 may be designed with a certain amount of elastic flexibility to pivot around a connection point 126, and the elasticity of the connection point 126 may provide the needed force to return the boot 114 back to a neutral position after the boot 114 has been pulled, in certain example embodiments, the boot 114 may include an additional biasing feature 300 for returning the boot 114.

[0089] In the example illustrated in FIG. 21, the biasing feature 300 includes a spring-based return feature 302. A pocket 140 defined within the yoke 96 may receive a return spring 146. The pocket 140 of the yoke 96 is configured to position the spring between the rear housings 90. A rear end 304 of the return spring 146 may be designed to abut a rear wall 306 of the pocket 140 within the yoke 96, and a front end 308 of the return spring 146 may abut an internal rear face 310 defined within the boot 114 to bias boot 114 toward the forward position of FIG. 7 when released by the user. When the user pulls boot 114 longitudinally away from the connector portions 112, the spring 146 is compressed. Spring 146 moves the boot 114 back to the rest position of FIG. 7 upon release by the user. The pocket 140 of the yoke 96 may be accessible through an opening such as the aperture 155 defined on the boot 114.

[0090] Referring now to FIGS. 22-23, another version of the biasing feature 300 may include integrally molded fingers 312 that are unitarily molded with the yoke 96 that are configured to interact with the boot 114 to return the boot 114 back to a neutral position. In the version of FIGS. 22-23, the flexible biasing fingers 312 are provided within a horizontal pocket 314 defined on the yoke 96. The fingers 312 define contact points 316 that can contact an internal rear

face 310 defined within the boot 114 to bias the boot 114 toward the forward position. FIG. 23 illustrates the fingers 312 in a flexed position.

[0091] FIG. 24 illustrates another example of flexible biasing fingers 320 that are integrally molded with the yoke 96. In the example illustrated in FIG. 24, the flexible fingers 320 are provided within a vertical pocket 322 positioned between the side slots 98 defined by the yoke 96. Similar to the example shown in FIGS. 22-23, the vertical flexible fingers 320 define contact points 324 that can contact the internal rear face 310 defined within the boot 114 to bias the boot 114 toward the forward position.

[0092] By providing the biasing features 300 such as the spring 146 in between the rear housings 90 or by eliminating a separate, removable biasing feature such as a spring and forming the biasing feature as an integral part of the yoke 96 via structures such as the flexible fingers 312, the overall length of the connector 110 can be reduced.

[0093] FIGS. 25-26 illustrate another example of a duplex connector 410 that has features similar to that of the connector 110 described above. In the example connector 410 shown in FIGS. 25-26, a yoke portion 496 of the connector 410 is designed for top loading the connector portions 112 instead of a side-loading design such as that illustrated for connector 110 described above.

[0094] As shown in FIG. 25, the yoke 496 defines side-by-side pockets 457 for receiving the portions of the rear housings 90 that include the square annular recess 253 and the rear shoulder 256 from thereabove. The yoke 496 further defines opposing flexible tabs 411 flanking each pocket 457 that are configured to capture the rear shoulder 256 of the rear housings 90 once the rear housings 90 have been inserted into the pockets 457. As shown, the internal configuration of the pockets 457 of the yoke 496 and the rear annular recess 253 defined by the rear housing 90, including the rear shoulder 256, intermate to rotationally fix the rear housing 90 with respect to the yoke 496. And, as noted above, the flexible tabs 411 retain the rear housings 90 within the pockets 457 in a vertical direction. Thus, similar to the embodiment of the connector 110 discussed previously, in the connector 410, when the front housing 132 is rotated with respect to the rear housing 90, the rear housing 90 and the yoke 496 (and the boot 114, which is rotationally fixed with respect to the yoke 496) stay stationary.

[0095] As further shown, a cable path 498 is defined within the yoke 496 for receiving the cables from the top of the yoke 496. The cable path 498 extends all the way from a crimp region 404 defined by the yoke 496 to the pockets 457, splitting into two separate sections 489 for leading to each of the pockets 457. As noted for the embodiment of the connector 110, the cable path 498 may be designed to guide the fibers directly back into the jacket of the cable and limit deflection or unwanted bending of the fibers if the ferrules at the front of the connector 410 are pressed and moved rearward.

[0096] In the example of the connector 410 shown in FIGS. 25 and 26, the crimp region 404 of the yoke 496 is similar in configuration to that of the yoke 96 shown in FIG. 15 in that the crimp region 404 is provided as a rear projection 499 that is integrally formed with and extends from the yoke 496. Similar to the crimp region 104, the crimp region 404 defines an exterior surface 497 that is textured (e.g., knurled, ridged, provided with small projections, etc.) to assist in retaining the strength members (e.g.,

aramid yarns) on the yoke 496. Similar to the crimp region 104 discussed above, in the depicted example of the crimp region 404, the exterior surface 497 defines a series of parallel rings or ridges 495 extending in a direction from the rear end of the rear projection 499 toward the main body of the yoke 496. As shown, some of the rings 495 may define discretely spaced scalloped portions 493 that are positioned peripherally around the rings 495. The scalloped portions 493 define teeth 401 thereinbetween. As noted in the previous examples, the scalloped portions 493 (and the teeth 401) may be provided in an offset or staggered relationship between adjacent rings 495 as the rings 495 extend front to back. In this manner, when the strength members are pressed underneath a crimp sleeve 106 against the exterior surface 497 of the rear projection 499, the strength members will tend to spread out and follow zig-zagging patterns as they are captured underneath the crimp sleeve 106, leading to increased yarn and jacket retention. It is also contemplated that the teeth 401 may be configured such that they are deformed or smashed against the strength members to provide further retention.

[0097] As shown in the bottom perspective view of the yoke 496 in FIG. 26, a pocket 403 may be defined by the main body portion of the yoke 496 for accommodating any excess strength members after the strength members have been crimped at the crimp region 404.

[0098] Referring now to FIG. 27, another version of a connector 510 that has features similar to connectors 110 and 410 discussed above is illustrated. The version of the connector 510 shown in FIG. 27 has a top-load yoke design similar to the connector 410 shown in FIGS. 25-26, and includes similar bend or deflection limiting features for the fibers. However, in the connector 510, the crimp region 504 is provided as a separate crimp body 505 that is removably mounted to a yoke 596 of the connector 510 instead of being integrally formed with the yoke as a rear extension. With the design illustrated in FIG. 27, the crimp body 505 can be used to crimp the strength members of the cables before the cables are terminated to the connector portions 112. Once cables have been crimped and terminated to the connector portions 112, the crimp body 505 can be slidably inserted into a complementary slot 503 defined on the yoke 596, while the rear shoulders 256 of the rear housings 90 are inserted into pockets 557 defined within the yoke 596.

[0099] Now referring to FIG. 28, another example of a connector 610 that has features similar to those of connectors 110, 410, and 510 is illustrated, including fiber bend limiting features within the yoke. However, in the connector 610 shown in FIG. 28, the rear housings and the yoke are combined into a single integral structure 690 forming the rear portion of the connector 610. Similar to connectors 110, 410, and 510 discussed above, the structure 690 defines a front extension 691 that has a front annular ring 651 and a rear annular ring 652 with an annular recess 650 therebetween. As described with respect to the connector 110 discussed above, flexible portions 254 of the sidewalls of the front housings 132 can fit within the annular recess 650 defined by the front extension 691 when each front housing 132 is coupled to the structure 690 of the connector 610. In this manner, the front housings 132 are rotatable relative to the structure 690 for polarity reversal, but the front housings 132 are axially fixed with respect to structure 690.

[0100] It should be noted that although the connector 610 is illustrated with a crimp region 604 that is provided as a

separate crimp body 605 that is removably mounted to the structure 690, in other examples, the crimp region can be integrally formed with the structure 690 as a rear extension as discussed for other above examples shown, for example, in FIGS. 15, 17-19, 25, and 26.

[0101] Referring now to FIG. 29, another version of a connector 710 that has features similar to connectors 410, 510, and 610 discussed above is illustrated. The version of the connector 710 shown in FIG. 29 has a top-load design similar to the connectors 410 and 510 shown in FIGS. 25-27. However, in the connector 710, a yoke 796 does not define flexible tabs/latches for retaining the rear shoulders 256 of the rear housings 90 within pockets 757 when the rear housings 90 are inserted into the yoke 796 from above. The rear housings 90 are simply captured against the yoke 796 with a snap-fit cover 705 as shown in FIG. 29. The cover 705 defines downwardly extending snaps 706 on each of its right and left sides that are configured to be coupled to hooks 707 defined on each side of the yoke 796. The cover 705 also defines a plurality of tabs 708 adjacent a front end of the cover 705 that fit into snaps 709 defined adjacent the front of the yoke 796 to retain the cover on the yoke 796.

[0102] As shown, the cover 705 captures against the yoke 796 a biasing member in the form of a spring 700 that is positioned within a center pocket 701 defined on the yoke 796. The cover 705 defines an aperture 711 for accommodating a portion of the spring 700. As discussed above, the spring 700 is configured to contact the boot 114 and provide a force to return the boot 114 back to a neutral position after the boot 114 has been pulled.

[0103] It should be noted that the cover 705 may also define protrusions at a bottom side thereof that are configured to keep and constrain the fibers 156, 158 within fiber pockets 703 defined in the yoke 796 and to limit deflection of the fibers, as discussed above for the previous embodiments.

[0104] Now referring to FIGS. 30-32, another example of a connector 810 having features that are examples of inventive aspects in accordance with the present disclosure is illustrated. The connector 810 is similar to the connector 610 shown in FIG. 28 in the sense that the rear housings and the yoke are molded integrally. However, the combined integral structure is formed from two halves 890, 892 that are configured to be coupled together in a side-by-side relationship. As will be discussed in further detail below, when the two halves 890, 892 are brought together, the two halves 890, 892 capture in between both a biasing member in the form of a spring 800 and strength members of the fiber optic cable that is terminated to the connector 810.

[0105] As shown in FIGS. 30-32, the two halves 890, 892 each include intermating hook 893 and snap 894 structures for snap-fitting the two halves 890, 892 together.

[0106] Slots 895 are provided on each of the halves 890, 892 to capture rear protrusions 896 of the biasing spring 800 in between the two halves 890, 892 when the halves are latched together.

[0107] Rear portions of the two halves 890, 892 are designed such that they capture the strength members of the fiber optic cable in the center, whereas the fibers from the cable are lead around the outside within side slots 898, similar to the embodiments discussed above.

[0108] In addition to the intermating hook 893 and snap 894 structures, the two halves 890, 892 also each include

interfitting posts **801** and notches **802** for coupling the two halves **890, 892** together in a side by side relationship.

[0109] Still referring to FIGS. **30-32**, as noted above, the rear ends of the two halves **890, 892** cooperatively define a crimp region **804** of the connector **810**. As shown, similar to the designs discussed above, the crimp region **804**, when formed, may define full annular rings **899** on an exterior surface **897**. However, in contrast to the above-described designs, the crimp region **804** may be designed to capture the strength members in between the two halves **890, 892** when the halves are brought together. For this purpose, as shown, the two halves each may define an internal textured surface **805** (e.g., knurled, ridged, provided with small projections, teeth, etc.) to assist in retaining the strength members (e.g., aramid yarns) in between the halves. In the depicted embodiment, the two halves **890, 892** are shown to have inter-fitting teeth **806** that are provided in a relatively offset relationship so as to inter-fit together and capture the strength members of the fiber optic cable in between the two halves **890, 892**.

[0110] Furthermore, as shown, the halves **890, 892** may also define a post **809** provided thereinbetween, in front of the crimp region **804**, for wrapping the strength members around and leading them back toward the inter-fitting teeth **806** to provide further strain relief.

[0111] FIG. **32** illustrates the two halves **890, 892** in an assembled configuration showing the intermating of the various parts.

[0112] FIG. **33** illustrates a top perspective view of a portion of yet another embodiment of a fiber optic connector **910** having features that are examples of inventive aspects in accordance with the present disclosure. The fiber optic connector **910** has features similar to the connector **610** of FIG. **28**.

[0113] As shown, similar to the connector **610**, in the connector **910**, the rear housings and the yoke are combined into a single integral structure **990** forming the rear portion of the connector **910**. Similar to connectors **110, 410, 510**, and **610** discussed above, the structure **990** defines a front extension **991** that has a front annular ring **951** and a rear annular ring **952** with an annular recess **950** therebetween. As described with respect to the connector **110** discussed above, flexible portions **254** of the sidewalls of the front housings **132** can fit within the annular recess **950** defined by the front extension **991** when each front housing **132** is coupled to the structure **990** of the connector **910**. In this manner, the front housings **132** are rotatable relative to the structure **990** for polarity reversal, but the front housings **132** are axially fixed with respect to structure **990**.

[0114] It should be noted that the connector **910** may define a crimp region **904** that is provided as a separate crimp body that is removably mounted to the structure **990** (such as that shown in FIG. **28**), or may define a crimp region that is also integrally formed with the structure **990** as a rear extension as discussed for other above examples shown, for example, in FIGS. **15, 17-19, 25**, and **26**.

[0115] The version of the connector **910** requires pre-insertion of the fibers into the structure **990** prior to termination of the fibers to the front housings **132**. And, as shown, a biasing member in the form of a spring **900** is positioned within a center pocket **901** defined by the structure **990**, where the spring **900** is configured to contact the boot **114** and provide a force to return the boot **114** back to a neutral position after the boot **114** has been pulled, as discussed for

previous embodiments. The version of the connector **910**, similar to the connector **610**, also requires duplex polishing during termination.

[0116] In the present disclosure, the duplex connectors **110, 410, 510, 610, 710, 810**, and **910** are depicted and described as polarity-switching type connectors. Even though the depicted duplex connectors **110, 410, 510, 610, 710, 810**, and **910** may include polarity-switching features, as noted above, the inventive latching and unlatching features and other inventive features such as the above-described crimp regions with staggered, scalloped ridges may be used on any duplex or simplex connector, with or without polarity-switching features.

[0117] Although in the foregoing description, terms such as “top,” “bottom,” “front,” “back,” “rear,” “right,” “left,” “upper,” “lower,” “forward,” and “rearward” may have been used for ease of description and illustration, no restriction is intended by such use of the terms. The connectors described herein can be used in any orientation, depending upon the desired application.

[0118] The above specification, examples and data provide a description of the inventive aspects of the disclosure. Many embodiments of the disclosure can be made without departing from the spirit and scope of the inventive aspects of the disclosure.

1. A telecommunications connector comprising:

at least one connector portion including a front housing portion coupled to a rear housing portion, wherein the front housing portion is rotatable relative to the rear housing portion about a longitudinal axis defined by the front housing portion, the front housing portion including a ferrule assembly defined by a ferrule and a ferrule hub that holds the ferrule, wherein the ferrule assembly is mounted to the front housing portion such that the ferrule hub and the ferrule held by the ferrule hub are rotationally fixed with respect to the front housing portion and rotate together with the front housing portion relative to the rear housing portion when the front housing portion is rotated relative to the rear housing portion such that an optical fiber terminated to the ferrule is rotatable with the front housing portion, wherein the ferrule hub defines flats on an exterior surface thereof that mate with mating surfaces of a hub mount defined within an interior of the front housing portion for rotationally fixing the ferrule hub and the ferrule with respect to the front housing portion.

2-6. (canceled)

7. A telecommunications connector according to claim 1, wherein the at least one connector portion includes a latch for latching the at least one connector portion to a fiber optic adapter, wherein the at least one connector portion includes a boot that is movable longitudinally relative to the rest of the at least one connector portion for interacting with the latch for de-latching the at least one connector portion from the fiber optic adapter.

8-12. (canceled)

13. A telecommunications connector according to claim 7, wherein the at least one connector portion is configured to be latched to an LC format fiber optic adapter.

14. A telecommunications connector according to claim 7, wherein the latch is pivotally connected to the at least one connector portion at a connection portion, wherein the connection portion is an intermediate connection portion positioned between a front end and a rear end of the latch.

15. A telecommunications connector according to claim 7, wherein the front housing portion and the boot define intermating structures to act as positive stops to stop rotation of the front housing portion with respect to the rear housing portion.

16. A telecommunications connector according to claim 15, wherein the intermating structures are respectively defined by the latch of the front housing portion and a slot on the boot.

17. A telecommunications connector according to claim 1, comprising:

two of the at least one connector portions provided in a side-by-side relationship, each connector portion including a front housing portion coupled to a rear housing portion, wherein each front housing portion is rotatable relative to the rear housing portion about a longitudinal axis defined by the front housing portion to change the polarity of the two connector portions, each front housing portion including a ferrule assembly defined by a ferrule and a ferrule hub that holds the ferrule, wherein the ferrule assembly is mounted to the front housing portion such that the ferrule hub and the ferrule held by the ferrule hub are rotationally fixed with respect to the front housing portion and rotate together with the front housing portion relative to the rear housing portion when the front housing portion is rotated relative to the rear housing portion.

18. A telecommunications connector according to claim 17, wherein the telecommunications connector is a duplex fiber optic connector and each ferrule terminates a separate optical fiber.

19-33. (canceled)

34. A telecommunications connector comprising:

a front housing portion coupled to a rear housing portion and a crimp region, wherein the front housing portion includes a ferrule and a latch, wherein the latch defines a front end and a rear end, and is pivotable relative to the rest of the front housing portion at a connection portion;

the crimp region configured for crimping at least a portion of a cable terminated by the ferrule, wherein the crimp region defines a plurality of spaced apart crimp rings stacked longitudinally, each crimp ring defining discretely spaced scalloped portions that are positioned peripherally around the crimp rings, wherein the scalloped portions between adjacent crimp rings are provided in a staggered position in a longitudinal direction defined by the telecommunications connector.

35. A telecommunications connector according to claim 34, wherein the scalloped portions define teeth therebetween that are configured to be deformed against strength members of a cable to be terminated by the ferrule.

36. A telecommunications connector according to claim 34, wherein the telecommunications connector is a fiber optic connector and the ferrule terminates an optical fiber.

37. A telecommunications connector according to claim 34, further comprising a cable terminated by the ferrule crimped to the crimp region.

38. A telecommunications connector according to claim 34, wherein the front housing portion is rotatable relative to the rear housing portion about a longitudinal axis defined by the front housing portion.

39. A telecommunications connector according to claim 38, further comprising a yoke coupled to the rear housing

portion, wherein the rear housing portion is rotationally and axially fixed with respect to the yoke, the crimp region of the telecommunications connector provided on the yoke.

40. A telecommunications connector according to claim 39, wherein the crimp region is integrally formed with the yoke.

41. A telecommunications connector according to claim 39, further comprising a boot mounted to the yoke, the boot movable longitudinally relative to the yoke, wherein the boot is configured to engage the rear end of the latch and cause the front end of the latch to pivot toward the ferrule of the front housing portion as the boot is moved away from the yoke.

42. (canceled)

43. A telecommunications connector comprising:

at least one connector portion including a front housing portion coupled to a rear housing portion, wherein the front housing portion is rotatable relative to the rear housing portion about a longitudinal axis defined by the front housing portion, the front housing portion including a ferrule assembly defined by a ferrule and a ferrule hub that holds the ferrule, wherein the at least one connector portion includes a latch, wherein the latch defines a front end and a rear end, and is pivotable relative to the rest of the at least one connector portion at a connection portion, wherein the ferrule assembly is mounted to the front housing portion such that the ferrule hub and the ferrule held by the ferrule hub are rotationally fixed with respect to the front housing portion and rotate together with the front housing portion relative to the rear housing portion when the front housing portion is rotated relative to the rear housing portion, wherein the ferrule hub defines flats on an exterior surface thereof that mate with mating surfaces of a hub mount defined within an interior of the front housing portion for rotationally fixing the ferrule hub and the ferrule with respect to the front housing portion; and

a boot mounted on the telecommunications connector, the boot movable longitudinally relative to the rear housing portion, wherein the boot is configured to interact with the latch and cause the latch to pivot relative to the rest of the at least one connector portion as the boot is moved relative to the rear housing portion.

44. A telecommunications connector comprising:

two connector portions provided in a side-by-side relationship, each connector portion including a front housing portion, wherein the front housing portions are coupled to a rear housing portion that is formed from two rear housing halves snap-fit together, wherein each front housing portion is rotatable relative to the rear housing portion about a longitudinal axis defined by the front housing portion to change the polarity of the two connector portions, each front housing portion including a ferrule assembly defined by a ferrule and a ferrule hub that holds the ferrule and a latch, wherein the latch defines a front end and a rear end, and is pivotable relative to the rest of the front housing portion at a connection portion, wherein the ferrule assembly is mounted to the front housing portion such that the ferrule hub and the ferrule held by the ferrule hub are rotationally fixed with respect to the front housing portion and rotate together with the front housing portion relative to the rear housing portion when the

front housing portion is rotated relative to the rear housing portion, wherein the ferrule hub defines flats on an exterior surface thereof that mate with mating surfaces of a hub mount defined within an interior of the front housing portion for rotationally fixing the ferrule hub and the ferrule with respect to the front housing portion; and

a boot mounted on the telecommunications connector, the boot movable longitudinally relative to the rear housing portion, wherein the boot is configured to engage at least a portion of the latch of each of the front housing portions and cause the latch to pivot relative to the rest of the front housing portions as the boot is moved relative to the rear housing portion.

45. A telecommunications connector according to claim **44**, further comprising a crimp region cooperatively defined by the two rear housing halves, the crimp region configured for crimping at least a portion of a cable terminated by the telecommunications connector, wherein the crimp region

defined by the two rear housing halves is configured such that the two rear housing halves capture strength members of the cable thereinbetween when the two rear housing halves are brought together.

46. A telecommunication connector according to claim **45**, wherein each of the two rear housing halves defines surface texturing for capturing the strength members of the cable in between the two rear housing halves when the two rear housing halves are brought together.

47. A telecommunications connector according to claim **46**, wherein the surface texturing is provided in the form of intermeshing teeth.

48. A telecommunications connector according to claim **45**, further comprising a post positioned between the two rear housing halves for wrapping strength members therearound before capturing the strength members between the two rear housing halves when the two rear housing halves are brought together.

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